



**Verified Carbon
Standard**

SCATEC RMI HYBRID CLUSTER

Scatec

Anthesis 

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Scatec RMI Hybrid Cluster (the "Project") which has multiple locations or project activity instances (but is not a grouped project) with(3) equivalent project activities located adjacent to each other on a single operational site, is a dispatchable power facility, comprising of solar photovoltaic power feeding into a battery energy storage system (BESS) technology and being dispatched through the hybrid power plant controller (HPPC) prior to feeding into the national grid at the Rooidam switching station located onsite. The Project is located approximately 20 km North-East of the town of Kenhardt in the Northern Cape province of South Africa and is located near Eskom's Nieuwehoop Substation and the adjacent Eskom Rooidam switching station. The Project has a Contracted Capacity of 150 MW at the Delivery Point as per the Power Purchase Agreements (PPAs) and comprises an area of around 880ha. The solar PV portion is split over three project activities located adjacent to one another in close proximity to the central BESS, Rooidam switching station and Operating infrastructure and forms a combined total installed capacity of approximately 540 MW peak (DC). The installed capacity for each solar PV portion is 180 MW (DC). The BESS has a combined energy capacity of 1140 MWh, and 225 MW dispatch capacity, at the Delivery Point. Excess power is used to charge the battery energy storage system (BESS) in order to supply dispatchable power to the grid for prolonged periods as specified in the PPA. The HPPC ensures that the plant complies with grid requirements at the Point of Interconnection (POI). It also dynamically allocates power generation and storage discharge among the different components (e.g., PV, BESS) to maximize the use of available renewable energy and avoid overloading the grid connection based on real-time and forecast resource availability.

The Project achieved its Commercial Operation Date (COD) with the signature of the Notice of Commencement of the Facility between the facility operating entity and the national power utility formally commencing the Power Purchase Agreement [1]. This was achieved for the first project activity (cluster) of the project on 19 November 2023. This is the start date of the project and its associated GHG reduction activity. From this date, the plant operated, with remaining two project activities (clusters) starting with the achievement of COD on 26 November 2023 and 9 December 2023 respectively. No other changes from the project implementation outlined in this document were experienced during this initial verification period. Financial close for the project was reached in July 2022 with construction commencing in July 2022 [2]. All Operations and Maintenance of the project is conducted by a subsidiary Scatec company at the start date.

Prior to implementing this project, electricity was supplied by the current fleet of power plants owned and operated by Eskom, the majority of which are fossil fuel based. The Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is an initiative of the South

African government aimed at increasing electricity capacity through private sector investment in solar photovoltaic and concentrated solar, onshore wind power, small hydro, landfill gas, biomass, and biogas [3]. Since coal is an emissions-intensive energy carrier, South Africa's economy is very emissions-intensive, especially when compared to other countries at a similar stage of development as measured by the Human Development Index [4]. South Africa's energy sector is by far the largest contributor to the country's emissions, contributing 94.7% to the country's predominant GHG, which is carbon dioxide (CO₂) [5].

On average the project activity is expected to reduce annual GHG emissions equivalent to 810,840 tCO₂e and 8,108 400 tCO₂e over the crediting period of 10 years.

The total GHG emission reductions generated in the monitoring period from 19-November-2023 to 30 June 2025, is 1,266,749 tCO₂e.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	19-November-2023 – 18-November-2033	VCS	Carbon Check (India) Private Ltd.	Ten years (10) non-renewable
Verification	19-November-2023 – 30-June-2025	VCS	Carbon Check (India) Private Ltd.	1.5 years

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	Scope: 01 - Energy (renewable / non-renewable)
Project activity type	Renewable Energy Project

1.4 Project Eligibility

1.4.1 General eligibility

The project is eligible under the scope of the VCS Program since the project activity meets the criteria listed in paragraph 2.1.1 of the VCS Standard, Version 4.7. Table 1 below describes the scope of the VCS program and the project's applicability, and Table 2 further elaborates on the project's eligibility.

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

Table 1: Scope of the VCS program

Elements	Justification of project applicability
1) The seven Kyoto Protocol greenhouse gases.	The project aims at reducing greenhouse gas (GHG) emissions by replacing fossil fuel energy supplied to the South African national grid, with clean renewable electricity.
2) Ozone-depleting substances (ODS).	Not applicable
3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Project activity applies VCS-approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ² which has been approved for use under the VCS Program.
4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Project activity applies VCS approved CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0) ³ .
5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	N/A

Table 2: Excluded Project Activities under the VCS v 4.7 Table 1

Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants. The exclusion does not apply to ocean energy (e.g., wave, tidal, salinity gradient, and ocean thermal energy conversion). For hydro projects, large scale means a maximum capacity of greater than 15MW, where maximum capacity is the installed/rated capacity or authorized capacity (as determined in the activity approval from the project regulator, government, or similar entity),	Not applicable to this project which is not a hydroelectric power plant.

² <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>
³ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

whichever is lower. Grid-connected means >50% of total generation is exported to a national or regional grid.	
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems (e.g., batteries). Grid-connected means >50% of total generation is exported to a national or regional grid. See the VCS Program Definitions for the full definition.	This is a grid-connected electricity generation activity using solar photovoltaic (PV) and BESS technology with a hybrid power controller. Hence the project is eligible and not excluded.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration. The exclusion does not apply to waste gas recovery or electricity generation using waste heat recovery outside of combined cycle applications (e.g., organic Rankine cycles).	Not applicable to this project which is not a waste heat recovery project.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass. The exclusion does not apply to gasification, pyrolysis, combusting biofuels, biogas, fractions of renewable biomass in refuse-derived fuels, agro/forest biomass residues in waste streams that are sent to landfills, CO₂ capture and storage from renewable biomass combustion, or thermal efficiency improvements (e.g., cook stoves).	Not applicable to this project which does combust biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel. The exclusion does not apply to the use of captured flare and/or vent gas or waste containing previously used petroleum products (e.g., used plastics, oils, lubricants).	Not applicable to this project which does not combust fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs). Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO₂e per year.	Not applicable to this project which does not replace electric lighting.

Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers. Large-scale means energy-efficient improvements with a maximum savings greater than 60 GWh/year or emission reduction greater than 60 kt CO _{2e} per year.	Not applicable to this project which does not install or replace electricity transmission lines or transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	Not applicable to this project which does not reduce HFC-23 emissions.

Pipeline listing and validation deadlines: The project activity was listed on the VERRA website in April 2025, to initiate the validation process. The opening meeting with the VVB took place once the project activity had been listed and showed the status of ‘Under Validation’ in the VCS Program Registry. Furthermore, the VVB did not complete validation until the 30-day public comment period had ended. As per para 3.8.1 of the VCS standard V4.7, non-AFOLU projects shall complete validation within two years of the project start date and this condition will be fulfilled by this project activity as the start date of the project activity is 19 November 2023 which is less than two years prior to the completion of validation.

The project activity applies CDM methodology ACM0002: Grid-connected electricity generation from renewable sources (Version 22.0), which has been approved by Verra for use under the VCS Programme for project activities seeking registration with them. ACM0002 is a large-scale methodology and has no capacity limit, and the Scatec RMI Hybrid Cluster is a multiple project activity project in one location and is not part of a grouped project, a larger project or activity.

1.4.2 AFOLU project eligibility

This section is not applicable, as the project is not an AFOLU project.

1.4.3 Transfer project eligibility

This section is not applicable, as the project is not being transferred from another GHG Program nor a CPA seeking registration under the VCS Program.

1.5 Project Design

☐ Single location or installation

- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

Grouped Project Design

This is not a Grouped project. Hence, this section is not applicable.

1.6 Project Proponent

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1.7 Other Entities Involved in the Project

There are no other entities involved.

1.8 Ownership

The three Scatec Project Proponents. have the legal right to own the Scatec RMI Hybrid Cluster facilities, which are verified through the following documents:

- Eskom's Power Purchase Agreements dated June 2022 which includes all details of the physical locations and property boundaries.
- The Scatec Project Proponents have agreements with subsidiaries of the parent company (Scatec ASA) for functions such as Operations & Maintenance, Development Work, EPC, Management Services and other shared services, including the operational monitoring and power management via the Global Operations Centre (GOC).
- NERSA Generation Licences dated July 2022.[6]
- The Operator has a legal agreement and right to use the land with land lease agreements for the duration of the project lifetime, which can be verified through the following documents provided by the project proponent:
 - Lease agreements exist for all clusters and operations. The lease period is for a minimum of 20 years from COD. The lease covers the footprint of the land –

Remaining Extent of the farm Onder Rugzeer Number 168; Division Kenhardt; Northern Cape [7] [8] [9].

- Anthesis and the Scatec PPs have a legal agreement in place for the joint ownership of a registered carbon project. [10]

1.9 Project Start Date

Project start date	19-November-2023
Justification	As per §3.8 of the VCS Standard, the start date “is the date on which the project began generating GHG emission reductions.” The project activity started generating GHG Emission removals on the Commercial Operations Date (COD) which was signed and confirmed as being 19 November 2023 [11]

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	19-November-2023 to 18-November-2033

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals (ERRs) of the project:

- ☐ < 300,000 tCO₂e/year (project)
☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
19-November-2023 to 31-December-2023	87 841

01-January-2024 to 31-December-2024	810 840
01-January-2025 to 31-December-2025	810 840
01-January-2026 to 31-December-2026	810 840
01-January-2027 to 31-December-2027	810 840
01-January-2028 to 31-December-2028	810 840
01-January-2029 to 31-December-2029	810 840
01-January-2030 to 31-December-2030	810 840
01-January-2031 to 31-December-2031	810 840
01-January-2032 to 31-December-2032	810 840
01-January-2033 to 18-November-2033	722 999
Total estimated ERRs during the first or fixed crediting period	8 108 400
Total number of years	10
Average annual ERRs	810 840

1.12 Description of the Project Activity

The Scatec RMI Hybrid Cluster project (the "Project") is a multiple project activity on a single site, dispatchable facility, comprising of solar photo-voltaic renewable energy generating technology feeding into a battery energy storage system (BESS), through hybrid power controller. The solar photo-voltaic components form a cluster around the central BESS and transformer switching station that feeds the national grid at the Rooidam switching station on site. The Project is located approximately 20 km North-East of the town of Kenhardt in the direction of Groblershoop in the !Kheis Local Municipality of the Northern Cape province of South Africa and is located near Eskom's Nieuwehoop Substation, which is connected to the Rooidam switching station. The Project has a Contracted Capacity of 150 MW (AC) at the Delivery Point (Rooidam switching station). The solar portion is spread over three project

activities located adjacent to one another and forming a combined total installed capacity of an approximately 540 MW peak (DC) on the site. The BESS has a combined energy capacity of 225 MW dispatchable capacity, at the Delivery Point. Excess power is used to charge the battery energy storage system (BESS) in order to supply dispatchable power to the grid for prolonged periods as specified in the PPA. The HPPC ensures that the plant complies with grid requirements at the Point of Interconnection (POI). It also dynamically allocates power generation and storage discharge among the different components (e.g., PV, BESS) to maximize the use of available renewable energy and avoid overloading the grid connection based on real-time and forecast resource availability.

A project owned 132/33kV switching step-up substation (Rooidam switching station) is located within the project footprint with a 85 MVA 132/33kV Step-up transformer and transformer bay. The isolators on the HV Site of the IPP Transformer are the location of the Main Metering Installation, the Point of Connection (POC) and where energy will be exported from the Project to the national grid at the Delivery Point via a HPPC. The interconnection works include a 132 kV switching station, located adjacent to the IPP Substation, connected to the 132 kV network.

Project or Facility Details

Project Name	Scatec RMI Hybrid Cluster
Contracted Capacity [MW]	150 MW (3 x 50MW equivalent project activities)
Design Auxiliary Consumption of the Facility [MW]	Solar, 0.72 MW. Battery, 2.25 MW
No. of PV clusters	3 clusters with 2 sections in each.
No. of PV panels	990 864
PV Module type	JAM72D30-545 (540, 545, 550 Wp)
Tracker system	PVH Axone Duo (single-axis) (55° East to 55° West) See Figure 1 below
Type of inverter manufacturer	Huawei SUN2000 200KTL
PV Inverter units	2232
PCS Inverter type	BYD PSC9567-3150
PCS Inverter units	180
Number of Transformers:	MV: 30 - 37X Hv:1Xtbd
Transformers Details	Make: Actom Description: 132Kv to 33Kv power transformer Rating: 60 – 80 MVA

Planned voltage connection level:	132 kV
Emergency auxiliary genset	13 gensets on site to power control circuit if electrical system is down 13 gensets on site ⁴
Substation and transformer	Actom (Rooidam Substation)
Connection point (i.e. the name of the substation or distribution line onto which the Facility is intended to connect):	Rooidam Switching Station – Eskom POC on the HV Side of the IPP owned Step - up Transformer at the IPP project site. Rooidam will have a 132 kV network that will then feed into the Eskom Nieuwehoop station nearby
Battery Energy Storage System (BESS)	BYD CubePro Solution Description: Integrated BESS utilizing Li-Ion Battery Modules, AC-coupled BESS system Rating: 75MW Power rating (Each Solar PV portion) and 225 MW for all three solar PV portions Usable Capacity: 380 MWh (Each Solar PV Portion) and 1140 MWh (Combined for all three solar PV portions.
Battery units	456
Hybrid Power Plant Controller	Inaccess 75MWp_380MWp MSSU (H-PPC &S-PPC)
PV Cleaning	PV modules will be cleaned robotically with mechanical brushes on a rolling schedule

This is a Greenfield project activity generating renewable electricity from the solar photovoltaic component feeding the BESS technology, managed by a hybrid power controller and supplying the electricity to the national grid. In the baseline scenario the equivalent of electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the TOOL 07 “Tool to calculate the emission factor for an electricity system”, version 7.0. There is no technology transfer occurring in the project activity. There was no project related activity on the site of the project owner prior to the implementation of this project. Hence pre-project scenario and baseline scenario are the same.

⁴ Gensets are only used for backup to power the facility and not to generate dispatchable power or feed BESS.

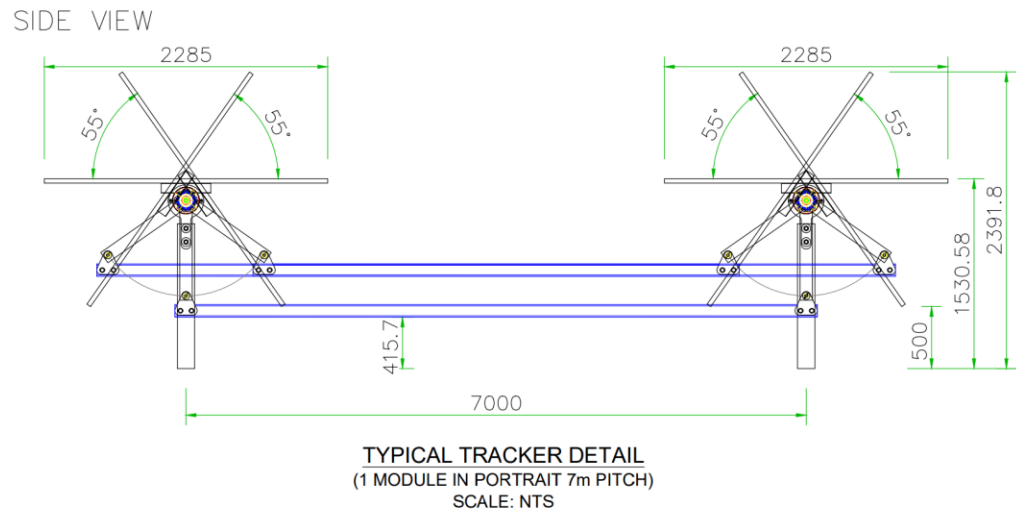


Figure 1: Typical PV Tracker system configuration

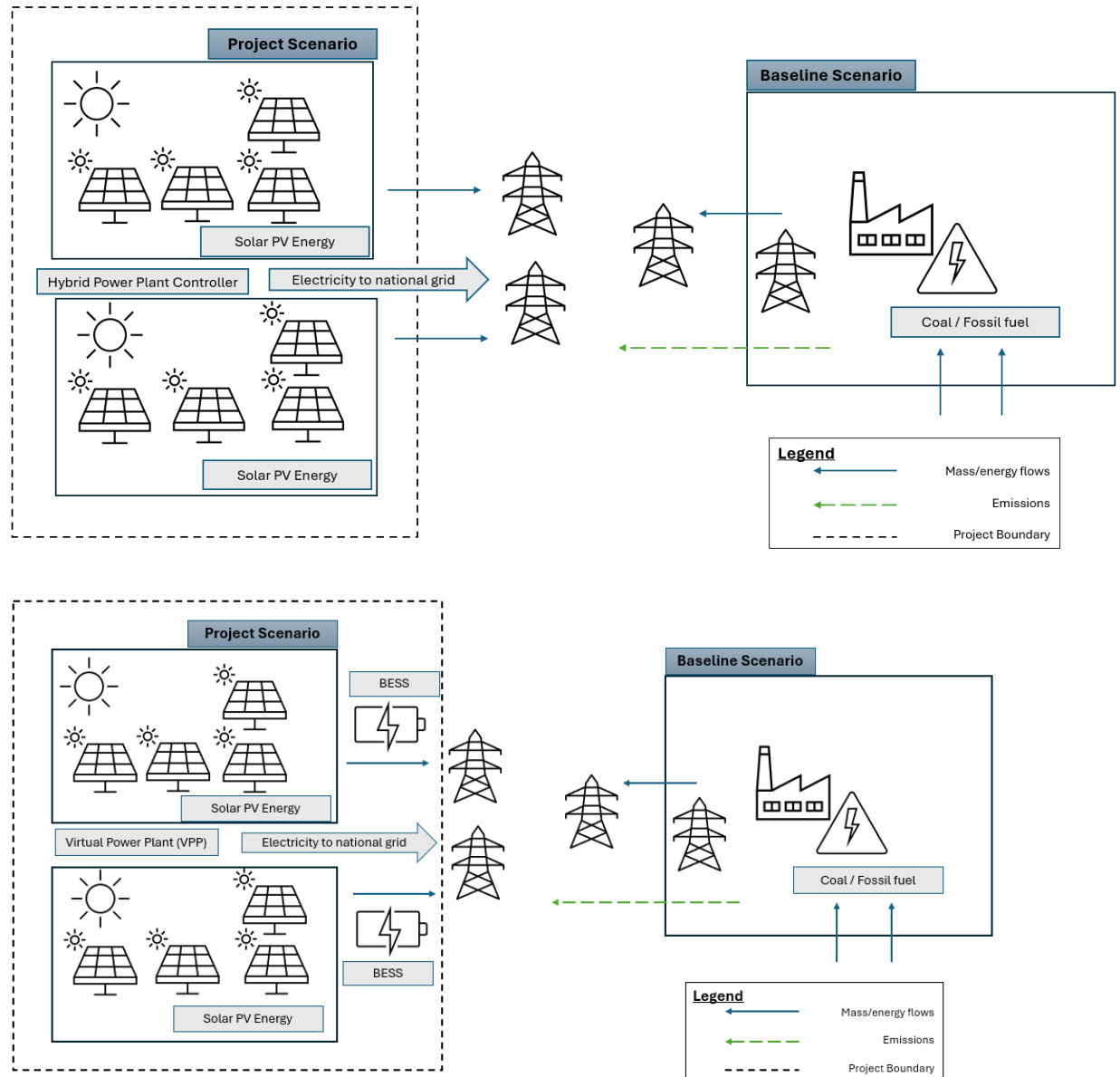


Figure 2: Schematic of project activities

1.13 Project Location

Project physical location and geographic coordinates

Physical address	Remaining Extent of the farm Onder Rugzeer Number 168; Division Kenhardt; Northern Cape
Latitude	29 ° 12' 19.55"S
Longitude	21 ° 18' 5.54"E

A KML file has been provided on the VERRA Registry that includes the area of the multiple project activities around this point.



Figure 3: Map showing project location

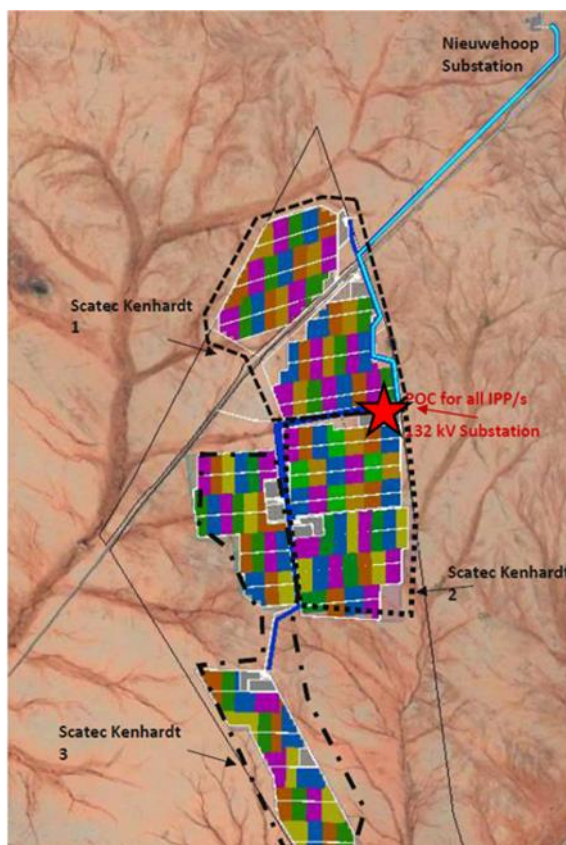


Figure 4: Physical location and layout of the multiple project activities on site

1.14 Conditions Prior to Project Initiation

Ecosystem type

The project site falls within the Nama Karoo biome which is characterized by semi-arid climatic conditions and vegetation that is dominated by shrubland. The ecosystem supports burrowing faunal species such as the *Suricata suricata*, and larger mammals such as the *Antidorcas marsupalis* which are not reliant upon the study area but are likely to forage over extensive ranges beyond the study area [12].

Current and historical land-use

The current land-use of the project site is a railway line known as the Sishen-Saldanha railway line which runs through the farm to the south of the project site with its associated infrastructure. The project site is located within a sheep farming agricultural region with no history of land cultivation on the farm. Due to the arid conditions and soil limitations, the site is not suitable for any agricultural land use other than low-intensity grazing. Present and prior environmental conditions of the project area:

Climate

The average annual temperature of the Kenhardt area is 19.6°C, with highs reaching 33°C in summer, and an annual average low temperature of 11°C in winter. The area receives very low annual rainfall levels of 183 mm that occur mainly during the autumn season. Since temperatures are still relatively high in autumn, the associated evaporation rates are high, leading to a semi-arid climate.

Topography

The region wherein the project site is located can be described as a generally level portion of land with a low gradient and low open hills. The low vegetation and shrubland produce a mottled background which is effective at making development types such as power lines and pylons blend in with the background. [12]

Hydrology

The project site lies across a low ridge that bisects the area into two watersheds. The area has many ephemeral drainage lines which feed a major water course: Wolfkopseloop. This water course feeds into the Hartebees River which flows into the Orange River to the north. The groundwater quality is described as poor due to the low groundwater recharge conditions as a result of low rainfall and high evaporation rates.

Soils

The soil type comprises of predominantly shallow, red sands to loamy sands that fall into the arid Silicic, Calcic and Lithic soil groups. The project site has low to moderate water erosion hazard, mainly due to the low slope, but it is susceptible to wind erosion due to the sandy texture of the soil.

Vegetation

The project site is classified as a Bushmanland Arid Grassland veld type, and is dominated by *Rhigozum* shrubs and small trees, as well as *Aristida* grasses. 99% of the vegetation in the site remains intact, therefore the conservation status of the vegetation type is classified as Least Threatened, with only two species of provincial conservation significance, namely the *Aloe dichotoma* and the *Aloe claviflora*.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Compliance with provincial, national and international legislation was considered during the planning and implementation of this project. The following legislative aspects were considered:

Table 3: Relevant Legislation

Constitution of the Republic of South Africa (Act 108 of 1996)	The Bill of Rights, in the Constitution of South Africa (No. 108 of 1996), gives everyone the
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	right to a non-threatening environment and requires that reasonable measures are applied to protect the environment. This protection encompasses preventing pollution and promoting conservation and environmentally sustainable development. These principles are embraced and given further expression in NEMA.
National Development Plan, Agenda 2030	The NDP envisages that, by 2030, South Africa will have an energy sector that provides reliable and efficient energy service at competitive rates, that is socially equitable through expanded access to energy at affordable tariffs and that is environmentally sustainable through reduced pollution.
Environmental Conservation Act (No. 73 of 1989)	The ECA provides for the effective protection and controlled utilisation of the environment and for matters incidental thereto.
National Environmental Management Act, (No. 107 of 1998)	NEMA requires adherence to the principles of Integrated Environmental Management (IEM) in order to ensure sustainable development, which, in turn, aims to ensure that environmental consequences of development proposals be understood and adequately considered during all stages of the project cycle and that negative aspects be resolved or mitigated, and positive aspects enhanced.
National Water Act (No. 36 of 1998)	Provides the legal framework for the effective and sustainable management of our water resources.
Conservation of Agricultural Resources Act (No. 43 of 1983)	CARA provides for control over the utilization of the natural agricultural resources in order to promote the conservation of the soil, water sources and vegetation and the combating of weeds and invader plants; and for matters connected therewith.
Occupational Health and Safety Act (No. 85 of 1993)	OHSA requires the employer to bring about and maintain, as far as reasonably

	practicable, a work environment that is safe and without risk to the health of the workers. This means that the employer must ensure that the workplace is free of hazardous substances, such as benzene, chlorine and micro-organisms, articles, equipment, processes, etc. that may cause injury, damage or disease. Where this is not possible, the employer must inform workers of these dangers, how they may be prevented, and how to work safely, and provide other protective measures for a safe workplace.
The Basic Conditions of Employment Act (No. 75 of 1997)	The BCEA intends to give effect to the right to fair labour practices referred to in section 23(1) of the Constitution by establishing and making provision for the regulation of basic conditions of employment; and thereby to comply with the obligations of the Republic as a member state of the International Labour Organisation; and to provide for matters connected therewith.
Electricity Regulation Act (Act 4 of 2006) and the Electricity Regulation on New Generation Capacity R399 of 4 May 2011, as amended by R1366 of 04 November 2016	The ERA establishes a national regulatory framework for the electricity supply industry. It designates the National Energy Regulator of South Africa (NERSA) as the custodian and enforcer of this framework. The act covers licensing, registration and regulation of electricity generation, transmission, distribution, trading, import and export. The Regulations provide for the procurement of new generation capacity derived by organs of state, from renewable energy sources and co-generation; base load, mid-merit load and peak load new generation capacity; and cross border projects. The South African Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is formulated under the framework of the Electricity Regulation Act, 2006 (Act No. 4 of

	2006) and the National Energy Act, 2008 (Act No. 34 of 2008), in conjunction with the Integrated Resource Plan (IRP). These acts provide the legal basis for the procurement of renewable energy and the establishment of a regulatory framework for the electricity supply industry, including the role of the National Energy Regulator of South Africa (NERSA).
Integrated Resource Plan (IRP2019), and the draft IRP 2023	The IRP is the electricity infrastructure development plan, of the Department of Energy, based on the least-cost electricity supply and demand balance, taking into account security of supply and the environment (minimize negative emissions and water usage). The South African Renewable Energy Independent Power Producer Procurement Programme (REIPPPP) is formulated under the framework of the Electricity Regulation Act, 2006 (Act No. 4 of 2006) and the National Energy Act, 2008 (Act No. 34 of 2008), in conjunction with the Integrated Resource Plan (IRP). These acts provide the legal basis for the procurement of renewable energy and the establishment of a regulatory framework for the electricity supply industry, including the role of the National Energy Regulator of South Africa (NERSA).
Carbon Tax Act 15 of 2019 and supporting regulations	Provide for VCUs to be used to offset carbon tax liability.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The sustainability objective of the project lies in its capacity to reduce greenhouse gas emissions. In alignment over the ten-year, non-renewable crediting period, the project anticipates an average annual emission reduction of 810,840 tons of CO₂ equivalent, totalling approximately 8,108 400 tons of CO₂ equivalent over the project lifetime. This substantial reduction aligns with Sustainable Development Goal (SDG) 13 (Climate Action), specifically addressing the urgent need to combat climate change and its impacts. Monitoring will be undertaken as per the plan in this Project Description, which involves the monitoring of generated renewable energy in MWh/year. This further aligns with South African SDG 7.2 which aims to increase the renewable energy within the country. The project displaces approximately 870,000 MWh per annum of emission-intensive grid electricity through the generation of low-emission renewable electricity.

Furthermore, aligned with SDG 8.5, the project will promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all. The operational phase will create 17,918.44 in people months across the three entities for the lifetime of the project. The metric for jobs created by the project is reported in people-months to capture both the number of people engaged and the duration of their work for a more accurate measure. The short-term employment opportunities accounted for in the construction phase of the project play a pivotal role in the job seekers' employability as they will acquire skills that will enhance their prospects for future job opportunities.

1.18.2 Sustainable Development Contributions Activity Monitoring

Table 3: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	7.2	7.2.1A1 (South African SDG Indicators) – Amount of renewable energy at annual operating capacity	Implemented activities will increase the amount of renewable energy in South Africa	Displaces approximately 870,000 MWh per annum of emission-intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 8,700 GWh of emission-intensive electricity through the generation of low emission renewable electricity over the project lifetime.
2)	13.0	Tons of greenhouse gas emissions avoided or removed	The total GHG emissions of the country will be reduced by the ERs and REMs that will no longer be contributing to the country's total GHG emissions. It is expected that the number of GHG emissions will be reduced annually as the project will be verified, confirming that these are no longer emitted to the atmosphere.	Displaces approximately 810,840 tCO _{2e} per annum of emission intensive grid electricity through the generation of low emission renewable electricity.	Displaces approximately 8,108,400 tCO _{2e} of emission-intensive grid electricity through the generation of low emission renewable electricity over the 10-year project lifetime.

3)	8.5	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	With total unemployment in South Africa having risen from 25.1% to 32.6% over the past decade (2014 to 2024) [13], this SDG target remains a high priority for the country. The country has a target to, “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.”	The project will provide 17,918.44 employment opportunities in people months per SPV for the 10-year operational period. During the operational period between Q1 2024 and Q1 2025, the project created an average of 1,481.39 direct jobs in people months, as tabulated below [14]. The employment provided includes skilled roles in technical operations and plant maintenance, as well as unskilled roles such as security and cleaning at the project sites. As per table below, actual jobs per quarter and year are presented [14]. <table><tr><th>Date</th><th>Actual jobs created (person months)</th></tr><tr><td>2024 Q1</td><td>1268.5</td></tr><tr><td>2024 Q2</td><td>1619.8</td></tr><tr><td>2024 Q3</td><td>1968.8</td></tr><tr><td>2024 Q4</td><td>1701.7</td></tr><tr><td>2025 Q1</td><td>2139.8</td></tr></table>	Date	Actual jobs created (person months)	2024 Q1	1268.5	2024 Q2	1619.8	2024 Q3	1968.8	2024 Q4	1701.7	2025 Q1	2139.8	The project planned to create approximately 17,918.44 (5,972.81 per cluster) jobs in people months across all the project sites during the operational phase. Over the 10-year operational period, a percentage of the revenue will be dedicated to local communities through socio-economic initiatives. These are monitored and reported upon by the project HR team.
Date	Actual jobs created (person months)																
2024 Q1	1268.5																
2024 Q2	1619.8																
2024 Q3	1968.8																
2024 Q4	1701.7																
2025 Q1	2139.8																

1.19 Additional Information Relevant to the Project

Leakage Management

As per the applied methodology ACM0002 (Ver. 22.0), no leakage emissions are considered in the project activity.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No further information submitted.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The approach adopted in the stakeholder identification process was to contact as many stakeholders as possible through several channels, which included:

- 1) Placing advertisements in a local newspaper, Die Gemsbok on the 18th of April 2025 [15].
1. Placing a noticeboard at the site and at strategic public locations in the vicinity of the project with a request for feedback. This included local supermarkets, municipality buildings and site access.
2. Providing written and email notice to the stakeholders on the register, including surrounding landowners, organs of state and relevant authorities; and
3. Requesting stakeholders to recommend other potential stakeholders to include them as members of the stakeholder database (chain referral process).

The stakeholder register is continuously updated as new I&APs are identified and/ or requested to be included or excluded in the database as part of the engagement process. The stakeholder register is maintained centrally.

All stakeholder details were compiled into a database which includes contact details, dates and details of consultations, and a record of all issues raised, and

	responses provided. This database will be updated on an on-going basis. Authorities regarded as key stakeholders: - Northern Cape Department of Environment and Nature Conservation (DEANC) - Department of Agriculture, Forestry and Fisheries: (DAFF): Directorate Land Use and Soil Management - Northern Cape: Department of Agriculture - Department of Water Affairs (DWA): Deputy Director Lower Orange Water Management Agency (WMA) - South African Heritage Resources Agency - Northern Cape Provincial Heritage: Boswa ya Kapa Bokone - ZF Mgcawu District Municipality - !Kheis Local Municipality - Kai !Garib Local Municipality and - Eskom
Legal or customary tenure/access rights	The legal tenure of the land is private ownership, and a binding lease agreement is in place with the landowners, in order to establish the Scatec RMI Hybrid Cluster plant facilities on the land. A binding lease agreement between the Project and the landowner was entered on the 6th of June 2022.
Stakeholder diversity and changes over time	The project site lies within the !Kheis Local Municipality, and key stakeholders that were identified include government agencies, neighbouring landowners and residents, local communities, businesses and environmental NGOs. The stakeholders cover a diverse and representative spectrum of all aspects, including race, gender and economic profile in the given area. The economic sector in the local municipality is mainly dominated by agriculture which provides 51.8% of jobs [16]

	however, poverty is prevalent in the area, with a recorded unemployment rate of 27.4%. The mining sector is the second-most dominant industry in the broader district municipality of ZF Mgcawu that provides work for local people. The two dominant industries provide male-oriented employment opportunities, which causes an uneven distribution of jobs between genders. The Scatec RMI Hybrid Custer Project has rolled out a training programme, providing skills that enable employment in the project for unskilled local community members.
Expected changes in well-being	There are no significant negative impacts anticipated on the well-being of stakeholders. The project's implementation has brought multiple positive impacts, such as improved local economic activity and job creation. Over the 10-year operational period, the local community will also benefit from a community trust with an equity share in the project life to benefit historically advantaged communities.
Location of stakeholders	The project site is located in the Groblershoop local community (!Kheis Local Municipality), and stakeholders that are most likely to be impacted by the project reside in the surrounding towns and communities of: - Kenhardt - !Kheis Local Municipality - Kai !Garib Local Municipality
Location of resources	The Scatec RMI Hybrid Cluster Project facility is located on the remaining extent of Onder Rugzeer Farm 168, with connection points to the substation on the remaining extent of Portion 3 of Gemsbok Bult Farm 120 within the !Kheis Local Municipality, Northern Cape Province.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	3 August 2015 – April 2025 (and ongoing)
Stakeholder engagement process	An Environmental Impact Assessment [16] was conducted for the project activities in 2015 which consisted of the Public Participation Process (PPP) where interested and affected parties were identified as stakeholders in the project. A database was collated through the PPP and formed part of the database for the VCS Stakeholder Engagement. Stakeholders were contacted primarily via email to inform them about the carbon project, prior to the quarterly stakeholder meeting on the 17th of March 2025. Meetings were held at various locations (including Kenhardt and Groblershoop). This provided stakeholders with an opportunity to raise any queries or concerns about the carbon project. The meeting was held in a culturally appropriate and inclusive manner, and communication was facilitated in the local languages: English, Afrikaans and Setswana. In addition to the emails and calls, a newspaper article was published in the local newspaper outlet – Die Gemsbok –on the 18th of April 2025, where stakeholders were encouraged to provide consent and comments to a dedicated email provided by the carbon project proponent. Notices were also placed in public areas inviting comments.
Consultation outcome	The stakeholders were adequately informed about the carbon project and the process to be followed for validation and verification of the project. The information was well received, and stakeholders showed support of the carbon project and its positive impacts for the communities which will be benefitting from it. There were no concerns received via the dedicated comments section, nor any issues raised during the in-person engagement at the quarterly meeting.

Ongoing communication	Ongoing communication will be via direct electronic communication to stakeholders recorded in the registered stakeholders' database, as well as notices placed in public locations in the surrounding communities. In addition, the comments section on the Anthesis website will remain active throughout the lifetime of the project. Furthermore, ongoing liaison and communication between the project owner and neighbouring communities is facilitated by a locally employed facilitator (CLO). Regular face-to-face meetings and monitoring visits will also be conducted by the project owner.
Stakeholder input	Inputs and concerns received from stakeholders during the EIA process were carefully considered and addressed in the EIA and the project implementation. No issues or comments received during the local stakeholder consultation meetings were directly related to the carbon finance or VCS validation and verification process, therefore no updates to the project design have been considered necessary. Stakeholder comments will continue to be monitored and considered for future revision to the project design, operation and implementation.



Figure 5: Article in "Gemsbok" – 18 April 2025

2.1.3 Free Prior and Informed Consent

Obtaining consent

There is a legally binding land lease between the project site's landowners and the operating project proponent. No IPs (indigenous peoples), LCs (local communities) or customary rights holders are impacted by the Scatec RMI Project activities.

Outcome of FPIC

The outcome of the process is that the Scatec RMI Cluster Hybrid facility is established on private land for a period of 20 years with the option to renew. No other land resources or communities have been negatively impacted.

2.1.4 Grievance Redress Procedure

Development process

The project proponent has developed a grievance mechanism to keep track of all grievances received from stakeholders. This consists of grievance forms and a grievance box located at the project site. In addition to this, stakeholders are given the platform to raise any concerns during the quarterly meetings held in their respective communities. A grievance register is kept to record all concerns and issues raised by the stakeholders via the communication channels that the project proponent has provided. Scatec also maintains a global grievance and whistleblower system that allows for anonymous raising of grievances that will be logged and handled on all issues raised (both internally and externally).

In order to address grievances in a timely manner, the project proponent has set up a Grievance Committee that processes all stakeholder grievances received from the Community Liaison Officers stationed in the affected communities, as well as those submitted via the grievance box or recorded during the quarterly meetings.

Anthesis has also provided a platform for any carbon project related grievances to be submitted via email or through a feedback form on the Anthesis or Scatec website.

Grievance redress procedure

Stakeholders have an ongoing opportunity to log into a dedicated feedback form on the Anthesis website in order to leave comments and engage with the project team. This website will be part of the ongoing and continuous engagement between the project and stakeholders. Comments received will be logged and taken into account in project operation and future verifications.

	Any grievances, complaints or conflicts from stakeholders would be captured via a dedicated feedback form on the Anthesis website, or from the other channels mentioned above. The Grievance Redress [17] process consists of: • Receiving the issue from a stakeholder • Assigning it to the relevant manager/team • Responding to the stakeholder (which may include telephonic, written or face to face communication) in an attempt to understand and resolve the issue in good faith • Recording the outcome of the interaction. • Grievances that cannot be resolved amicably in this manner will be referred to mediation by a neutral third party and failing resolution shall be raised to either arbitration or courts of law in South Africa.
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Grievances received	Resolution and outcome
Lack of employment opportunities for women during the construction phase.	A meeting was held to address the grievance, and the project proponent representative explained that although the construction phase had labour intensive job opportunities, more than 20% of the workforce on the project were women. The resolution was to identify more opportunities where female workers would be recruited during the operational phase.
Unfair distribution of job opportunities across the municipalities impacted by the project	The project proponents explained the recruitment process to the stakeholders and members of the !Kheis Local Municipality, detailing the closer proximity of the other municipalities which led to the uneven distribution of job opportunities.

2.1.5 Public Comments

Following the Verra Global Stakeholder Comment (GSC) period, 4 anonymous comments were received via Verra.

Comments received	Actions taken
Comment 1 – Anonymous – submitted 16 May 2025 It has recently come to our attention that several projects listed below have recently completed their public stakeholder consultation, along the process of applying for registration to generate carbon credits under the VCS. 1. Scatec RMI Hybrid Cluster (https://registry.terra.org/app/projectDetail/VCS/5488) 2. De Aar Hydra Storage (https://registry.terra.org/app/projectDetail/VCS/5337) 3. Oya Hybrid Project (https://registry.terra.org/app/projectDetail/VCS/5312) 4. Umoyilanga Hybrid Project (https://registry.terra.org/app/projectDetail/VCS/5272) 5. Redstone CSP project (https://registry.terra.org/app/projectDetail/VCS/5146) We have not received any notification from any of the project developers of their intention to pursue registration. As a significantly interested and affected stakeholder, we therefore object to the registration and request further consultation on several issues of concern.	Stakeholder engagement for the project followed the requirements stipulated in §3.18 of the VCS Standard. This included building on the extensive stakeholder process undertaken during the EIA phase of the project and augmenting this through a renewed review of the stakeholder database to identify and add to it, confirming any stakeholders with legal or customary rights tenure or access rights or any conflicting rights. The stakeholder engagement process included physical notices located in public areas (municipal buildings, local supermarkets etc.) near the project activity, meetings with local stakeholders, advertisements in newspapers, electronic (email) communication, some telephonic calls and a dedicated website link is maintained (https://www.anthesisgroup.com/za/project-comments/) that remains open for stakeholder concerns, complaints, grievances and engagement. As part of the operational phase, each project also has a dedicated team of specialists located on site specifically focused on stakeholder engagement. A 30-day Global Stakeholder Consultation (GSC) managed on the Verra registry website was also part of the process as per §3.18.9 – 3.18.13 of the stakeholder engagement process. The stakeholder engagement process was validated and checked by the

	independent VVB during the validation process, and included verification of documentation and communication, interviews with the project team members who were involved in the implementation and interviews with a sample of stakeholders themselves. The comments received from VERRA were anonymised, so it was unsure who the stakeholder was in order to review any possible comments or concerns raised during the engagement process. As part of our ongoing stakeholder engagement (as per §3.18.5 – 3.18.6 of VCS Standard) stakeholder may engage, comment, raise a grievance or be added to the stakeholder database. Various mechanisms are available, the primary ones being through the project operational stakeholder teams or via the weblink (https://www.anthesisgroup.com/za/project-comments/). While the stakeholder engagement process sought to identify and include all relevant stakeholders, any stakeholders missed, still have the opportunity to engage and be part of the ongoing process by accessing the weblink above.
Comment 2 – Anonymous – submitted 17 May 2025 Projects 5488, 5337, and 5146 have all been awarded tenders under South Africa’s REIPPPP and were procured by Eskom through government-backed bidding rounds. The tariff structure is designed to ensure the bankability and long-term viability of the projects. These are large-scale renewable energy projects with significant credit generation potential. We reached out to Eskom and confirmed that the renewable energy from these	The response to the comment has been split into the primary issues raised as follows: • Additionality • Ownership of project/Vcus • Potential for double or triple counting • Reflection in national Grid Emission Factor

projects is counted toward the national grid emission intensity. This raises several concerns, including; additionality, ownership, double or triple counting - reflected in the national grid factor, at least one project is selling electricity to a steel manufacturer that is using the green power to reduce its emissions intensity, raising the possibility of multiple claims on the same environmental benefits. There's also a stakeholder engagement concern - one project listed Eskom as having been engaged in April, but Eskom later indicated they were unaware of any such engagement. There could also be potential (non-public) legal disputes related to the projects.

- Potential selling green power to a non-Eskom offtaker
- Engagement with Eskom as a stakeholder
- Potential for (non-public) legal disputes

ADDITIONALITY: In accordance with the VCS Standard (§3.14) and the applied methodology (§5.2.3 and the associated Tools that were applied), additionality was evaluated within the framework applicable to the Verra VCS requirements, including CDM TOOL1 v7, CDM TOOL 24 v3.1 and CDM TOOL 27 v14. All parameters used in the assessment of additionality were valid at the time of investment decision, and were verified by the carbon project developer as well as the independent VVB before being submitted to Verra as part of the Registration process.

Even though the REIPPPP is a program implemented by the South African government, it is not a government subsidised program, but rather a price-competitive procurement programme. We acknowledge that REIPPPP tariffs are structured to provide cost recovery. However, the assumption that all awarded projects are fully bankable without carbon revenue is not borne out in practice. Of the projects commented on, only the Redstone CSP project is a REIPPPP project. The other projects, including this one, were all granted licenses under the Risk Mitigation IPP Programme. Despite this subtle

difference, the projects still had to demonstrate additionality in compliance with the Verra rules, irrespective if they were part of the REIPPP programme or not.

Additionally, risks such as capex overruns, long development lead times, logistics issues at remote greenfield sites and currency (ZAR) devaluation and volatility impacted financial viability, reinforcing the importance of a carbon revenue stream.

As such, the project was evaluated against the Verra requirements and found by the VVB to meet the criteria for additionality as per VCS rules.

Ownership of project/Vcus: In accordance with the VCS Standard (§3.7), project proponents need to demonstrate the legal right to control and operate the project activities. This and the proof of right were demonstrated to the VVB during the validation through legally binding agreements that the various project proponents had signed. These agreements covered commercial aspects of the project, including responsibilities for various project activities (such as operations, offtake, carbon development etc.).

While the comment suggests a contractual obligation limiting the ownership of the carbon credits to the project proponent, independent legal opinion and review obtained by both project proponents has a differing opinion in this, as the relevant clauses

of both the PPA and the Implementation Agreements (IA) are ambiguous regarding the ownership. Where a contract is ambiguous, then one can rely on general principles of law to interpret the contract, and in this instance reliance can be placed on the general principle of property law that the (carbon) project owner (the IPP) is also owner of the environmental attributes with the right to monetise such environmental attributes. Legal review was conducted in the light of these comments and a letter to the IPP from the IPPO dated 1 July 2025. The review was unable to identify any substantive contractual foundation that differs from the PP's current understanding and interpretation of the contracts and the eligibility to register a project and gain VCU's from the project.

Furthermore, these contractual obligations also made provision for changes to the operating environment in the future that may arise from changes in ownership, legal disputes, legislation, contractual amendments, policy changes or changes to the international voluntary carbon market, which the PP acknowledges may change any future claim or eligibility. These documents formed part of the independent validation by the VVB during the audit process.

Potential for double or triple counting: In accordance with the VCS Standard (§3.23), project proponents need to demonstrate that there is no double counting, or double selling of any GHG

emission reductions. This was confirmed by cross referencing a number of local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owner provided a signed declaration that no other environmental benefits are sought, and no offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process.

Reflection in national Grid Emission

Factor: Under the VCM, South Africa does not have a published and approved national grid emission factor or standardised baseline, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. CDM TOOL07 were applied in order to determine the grid emission factor. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects.

As per the prescribed tools, the projects seeking registration with Verra, as well as other REIPPP projects not yet commissioned at the time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the effect of other large scale

renewable energy projects already feeding into South Africa's grid at the time of publishing the PDD was taken into account through the Build Margin (BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements.

Potential selling green power to a non-Eskom offtaker: This had been checked during the project design and was confirmed again when addressing these comments. No evidence was found to substantiate this, and in line with the point above on double counting it was also validated by the VVB.

VVB has validated that the GHG emission reductions claimed by the project are based solely on the electricity produced and sold to Eskom in accordance with the PPA. Verification for emission reductions claimed is based on the calibrated meters measuring electricity that is fed to the national grid and there is no additional grid or feed that could be used for an additional offtake (wheeling) agreement or direct feed to an independent offtaker. The sites are also remote and have little potential to feed adjacent plants or municipalities.

Engagement with Eskom as a stakeholder: During the stakeholder identification process, Eskom had been

identified as a key stakeholder and representatives of Eskom had been included on the stakeholder register and contacted via email. They were provided with stakeholder updates and feedback was requested. While it is acknowledged that Eskom is a large organization, the project proponent is open to engage with more than one Eskom representative or department should they be added to the stakeholder register.

The stakeholder engagement process was validated and evidence and source documentation were verified by the independent VVB during the validation process.

Potential for (non-public) legal disputes:

It is acknowledged by the project proponents that legal disputes may arise with the project in the future for a number of reasons. While this comment is rather widely framed, it is acknowledged that a number of risks existing in these (and most other projects) such as Force Majeure, Liability and Indemnity clauses, penalty clauses, third party obligations and changes to governing frameworks (including legislation, regulations, policy or the international voluntary carbon market requirements) or the amendment of contracts.

Note the following acronyms are used:

- **REIPPPP:** Renewable Energy Independent Power Producer Procurement Programme

	• RMI: Risk Mitigation Independent Power Producer Procurement Programme • PPA: Power Purchase Agreement (concluded between Eskom and an IPP) • IA: Implementation Agreement (concluded between the DMRE and/or its predecessor or successor government department and the IPP) • IPPO: Independent Power Producer Office (an office within the Department of Mineral Resources and Energy) • DMRE: Department of Mineral Resources and Energy • IPP: Independent Power Producer (a project proponent)
Comment 3 – Anonymous – submitted 20 May 2025 PUBLIC COMMENTARY PERIOD: PROJECT VCS 5488 Clarity is sought regarding the potential to generate carbon offsets from projects registered under the South African Renewable Energy Independent Power Producer Procurement Programme (REIPPPP). It is noted project Scatec RMI Hybrid Cluster (5488) , as well as projects De Aar Hydra Storage (5337) and Redstone CSP Project (5146) have been awarded contracts under this national renewable energy procurement programme. 1. CONTEXT The REIPPPP is a competitive tender programme established to attract private sector investment into grid-connected renewable energy generation in South Africa. The programme is playing a critical role in driving the country's transition to cleaner	The response to the comment has been split into the primary issues raised as follows: • Clarity regarding potential to generate carbon credits • Management of carbon-related environmental attributes to avoid double counting • Environmental attributes already embedded in the GEF • If national utility (Eskom) acknowledges the rights to monetise carbon credits • Alignment with carbon accounting principles (related to GEF)

energy. Additional context on the REIPPPP is provided in Box 1.

Box 1: Overview of the REIPPPP and key stakeholders

The request aims to gain a deeper understanding of the eligibility of projects under the REIPPPP for the generation of carbon credits. Specifically, how environmental integrity is upheld and whether financial structures comply with international carbon market principles.

Clarity is being sought on two principal aspects:

- The management of carbon-related environmental attributes to avoid double counting, and
- The financial structuring of REIPPPP tariffs in relation to the principle of additionality.

2.1. Management of Environmental Attributes and Avoidance of Double Counting

It is understood that renewable energy generated through REIPPPP is included in the calculation of South Africa's national grid emission factor (GEF), which in turn is used to estimate the emissions attributable to grid-supplied electricity. The implication is that renewable electricity supplied to the grid contributes to a lower overall GHG intensity, thereby benefiting all grid users.

There is therefore a concern that issuing carbon credits for the same environmental benefit, already embedded in a reduced GEF, could constitute a form of double counting. To support a credible carbon market approach, clarity is needed on how the environmental attributes are treated contractually and institutionally.

Specifically, the following aspects require investigation:

- Financial structuring of REIPPPP tariffs within additionality
- If PPA explicitly defines ownership and transferability of the associated environmental rights
- Alignment with carbon accounting principles (related to REIPP projects already included in GEF)

Clarity regarding potential to generate carbon credits: REIPPPP tariffs are

structured for cost recovery and to ensure a reasonable return to investors. However, that does not preclude additionality in carbon market terms. Financial viability and additionality are not binary concepts — additionality is assessed in relation to whether the carbon revenue played a role in project decision-making or was necessary to overcome barriers (financial, technological, regulatory, etc.) at the time of investment.

Hence, not all REIPPPP projects are equally profitable or de-risked. Many face unexpected delays (e.g. grid constraints, currency issues, escalating capex, lead times in remote areas) or thin margins. These issues are material to an assessment of additionality.

So in principle these projects can qualify as carbon projects under the voluntary carbon market (as has already been demonstrated with a number of other REIPPPP projects in South Africa already registered under recognized carbon schemes such as Verra VCS or the CDM.

- Whether the Power Purchase Agreement (PPA) explicitly defines the ownership and transferability of the associated environmental rights.
- Whether Eskom, as the purchaser of electricity, acknowledges that the IPP retains the rights to monetise environmental attributes such as carbon credits.
- How the programme ensures alignment with carbon accounting principles, given that GHG reductions from REIPPPP projects are already reflected in the national GEF used for greenhouse gas accounting by electricity consumers.

2.2. Alignment of Tariff Structures with the Principle of Additionality

The principle of additionality is foundational to the issuance of carbon credits. In the case of REIPPPP, awarded tariffs are designed to ensure financial viability and bankability. The Power Purchase Agreement allows for cost recovery, typically based on the plant's Levelised Cost of Electricity, which includes capital investment, the weighted average cost of capital, capacity factor, and both fixed and variable operating and maintenance costs.

Given this, there is a concern that the inclusion of carbon credit revenue could be considered surplus to financial requirements, thereby potentially undermining additionality. If projects are already financially viable based on tariff revenues structured to cover all costs and provide a reasonable return on investment, the addition of carbon credit income might be perceived as a windfall, rather than a necessity for project feasibility.

It would be useful to understand if the programme has conducted any assessments or established guidelines to determine if carbon revenues are

The generation and use of carbon offsets for South African domestic purposes – including under the Carbon Tax Act (15 of 2019) and its associated offset regulations – have been explicitly recognized and encouraged by the National Treasury and South African Revenue Services. This project is aligned with this objective and the resultant VCUs (carbon credits) from these projects qualify under the domestic offset scheme and it is expected that they will be used within this framework. They are intended to support domestic decarbonisation and net zero targets, and as such directly support South Africa's NDC and industrial transition.

Management of carbon related environmental attributes to avoid double counting: In accordance with the VCS Standard (§3.23), project proponents need to demonstrate that there is no double counting, or double selling of any GHG emission reductions. This was confirmed by cross referencing a number of local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owned provided a signed declaration that no other environmental benefits are sought, and no offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process. No additional contractual obligations or

essential to the financial case for the project, or whether these are supplementary.

3. CONCLUSION

In closing, further insight into how the REIPPPP framework aligns with the requirements for carbon credit generation under international market standards would be appreciated. Specifically, if there are any applicable policy or regulatory provisions that may affect project eligibility, and whether certain bidding windows or project categories are treated differently. Clear guidance on these points will assist in evaluating the role of REIPPPP projects within credible carbon market mechanisms and support efforts to ensure environmental integrity and financial rigour in project development,

evidence of any carbon credits being claimed or accounted for by any other third party was found during this review.

In addition, during the project development process, each project owner provided a signed declaration that no other offtake agreements or GHG benefits have been allocated outside of the listed Verra project.

Environmental attributes already embedded in GEF: Under the VCM, South Africa does not have a published and/or approved national grid emission factor (GEF) or standardised baseline for the use by project proponents, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. The associated tool, CDM TOOL07 was applied in order to determine the GEF at the time of validation. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects.

As per the prescribed tools, all non-operational projects, e.g. those seeking registration with Verra, as well as other REIPPPP projects not yet commissioned at the time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the effect of all other large scale renewable energy projects already feeding into South Africa's grid at the

time of publishing the PDD were taken into account through the Build Margin (BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements.

If national utility (Eskom) acknowledges the rights to monetise carbon credits:

The specific rights to monetise carbon credits does not directly form part of the VERRA registration process, however project proponents do need to demonstrate ownership of any VCUs. This was responded to in the comment above and were demonstrated to the VVB during the validation through legally binding agreements that the various project proponents had signed. These agreements covered commercial aspects of the project, including responsibilities for various project activities (such as operations, offtake, carbon development etc.).

While the comment suggests a contractual obligation limiting the monetisation of the carbon credits to the project proponent, independent legal opinion and review obtained by both project proponents has a differing opinion in this, as the relevant clauses of both the PPA and the Implementation Agreements (IA) are ambiguous regarding the rights to monetise any environmental attributes. The environmental attributes are not clearly defined either in the PPA or IA. For

example, for the Scatec RMI Hbrid Cluster project, the PPA implicitly acknowledges the IPP's right to utilise carbon credits under the RMI. This implicit acknowledgement lies in the calculation of the IPP's cost recovery of carbon taxation payments made by the IPP on behalf of Eskom, which must be reduced by the maximum carbon offset permissible, whether the IPP actually earns any offsets, and thus implying that the value of such carbon offset is for the account of the IPP. Where a contract is ambiguous then one can rely on general principles of law to interpret the contract and, in this instance such reliance can be placed on the general principle of property law that the (carbon) project owner (the IPP) is also owner of the environmental attributes with the right to monetise such environmental attributes. Legal review was conducted in the light of these comments and a letter to the IPP from the IPPO dated 1 July 2025. The review was unable to identify any substantive contractual foundation that differs from the IPP's current understanding and interpretation of the contracts and the eligibility to register a project and gain VCUs from the project.

Furthermore, these contractual obligations also provide for changes to the operating environment in the future that may arise from changes in ownership, legal disputes, legislation, contractual amendments, policy changes and/or changes to the international voluntary carbon market, which the PP acknowledges may change any future claim or eligibility. These

documents formed part of the independent validation by the VVB during the audit process .

In addition to the above, the monetization of carbon credits resultant from a VCS project, would fall outside of the registration and issuance process, as this would be managed in terms of a commercial agreement between project proponents and shareholders regarding any revenue sharing.

Alignment with carbon accounting principles (related to GEF): Under the VCM, South Africa does not have a published and approved national grid emission factor or standardised baseline, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. CDM TOOL07 were applied in order to determine the grid emission factor. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects.

As per the prescribed tools, the projects seeking registration with Verra, as well as other REIPPP projects not yet commissioned at the time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the effect of other large scale renewable energy projects already feeding into South Africa's grid at the time of publishing the PDD was taken into account through the Build Margin

(BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements.

Financial structuring of REIPPPP tariffs within additionality: In accordance with the VCS Standard (§3.14) and the applied methodology (§5.2.3 and the associated Tools that were applied), additionality was evaluated within the framework applicable to the Verra VCS requirements, including CDM TOOL1 v7, CDM TOOL 24 v3.1 and CDM TOOL 27 v14. All parameters used in the assessment of additionality were valid at the time of investment decision, and were verified by the carbon project developer as well as the independent VVB before being submitted to Verra as part of the Registration process.

Even though the REIPPPP is a program implemented by the South African government, it is not a government subsidised program, but rather a price-competitive procurement programme. We acknowledge that REIPPPP tariffs are structured to provide cost recovery. However, the assumption that all awarded projects are fully bankable without carbon revenue is not borne out in practice. Of the projects commented on, only the Redstone CSP project is a REIPPPP project. The other projects, including this one, were all granted licenses under the Risk Mitigation IPP Programme. Despite this subtle

difference, the projects still had to demonstrate additionality in compliance with the Verra rules, irrespective if they were part of the REIPPP programme or not.

Additionally, risks such as capex overruns, long development lead times, logistics issues at remote greenfield sites and currency (ZAR) devaluation and volatility impacted financial viability, reinforcing the importance of a carbon revenue stream.

As such, the project was evaluated against the Verra requirements and found by the VVB to meet the criteria for additionality as per VCS rules.

If PPA explicitly defines ownership and transferability of the associated environmental rights: In accordance with the VCS Standard (§3.7), project proponents need to demonstrate the legal right to own, control and operate the project activities. This and the proof of right were demonstrated to the VVB during the validation through legally binding agreements between that the various project proponents had signed. These agreements covered commercial aspects of the project, including responsibilities for various project activities (such as operations, offtake, carbon development etc.).

While the comment suggests a contractual obligation limiting the ownership and transferability of the carbon credits to the project proponent, independent legal opinion and review obtained by both project proponents has

a differing opinion in this, as the relevant clauses of both the PPA and the Implementation Agreements (IA) are ambiguous regarding the ownership. Where a contract is ambiguous then one can rely on general principles of law to interpret the contract, and in this instance reliance can be placed on the general principle of property law that the (carbon) project owner (the IPP) is also owner of the environmental attributes with the right to monetise such environmental attributes. Legal review was conducted in the light of these comments and a letter to the IPP from the IPPO dated 1 July 2025. The review was unable to identify any substantive contractual foundation that differs from the IPP current understanding and interpretation of the contracts and the eligibility to register a project and gain VCUs from the project.

Furthermore, these contractual obligations also made provision for changes to the operating environment in the future that may arise from changes in ownership, legal disputes, legislation, contractual amendments, policy changes and/or changes to the international voluntary carbon market, which the PP acknowledges may change any future claim or eligibility. These documents formed part of the independent validation by the VVB during the audit process .

Alignment with carbon accounting principles (related to REIPP projects already included in GEF): Under the VCM, South Africa does not have a

	published and approved national grid emission factor or standardised baseline, and each project has to determine the Grid Emission Factor in accordance with the VCS Standard (§3.13) and the applied methodology. In the case of this project, CDM methodology ACM 0002 (v 22.0) was applied, and as per §5.2. CDM TOOL07 were applied in order to determine the grid emission factor. This process is approved by Verra and takes cognisance of all components of the national grid, including renewable energy projects. As per the prescribed tools, the projects seeking registration with Verra, as well as other REIPPP projects not yet commissioned at the time of publishing the PDD are not included in the baseline scenario and hence are excluded from the grid emission factor calculation. However, the effect of other large scale renewable energy projects already feeding into South Africa's grid at the time of publishing the PDD was taken into account through the Build Margin (BM) calculation. The build margin refers to the group of prospective power plants whose construction and future operation would be affected by the proposed project activity. This was demonstrated to the VVB during the validation and met the requirements.
Comment 4 – Anonymous – submitted 10 June 2025 FORMAL CONCERN REGARDING REGISTRATION OF IPP PROJECTS UNDER THE VERRA VCS PROGRAMME	The response to the comment has been split into the primary issues raised as follows: Carbon credits are subject to contractual provisions (§4)

1. We refer to the above-subject matter.

2. It has come to our attention that the following Independent Power Producer (IPPs) Projects have been registered or started the process of registering under the Verra Verified Carbon Standard (VCS) programme:

2.1 Scatec RMI Hybrid Cluster (VCS 5488)

2.2 De Aar Hydra Storage (VCS 5337)

2.3 Oya Hybrid Project (VCS 5312)

2.4 Umoyilanga Hybrid Project (VCS 5272)

2.5 Redstone CSP Project (VCS 5146)

3. These projects are being implemented under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP) and Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), managed by the Department of Electricity and Energy through the IPP Office.

4. We wish to formally inform Verra that, in accordance with the Power Purchase Agreements (PPAs), Implementation Agreements (IAs), and associated procurement documents governing these programmes, the ownership and treatment of environmental attributes, including carbon credits and associated claims, are subject to specific contractual provisions.

5. Under these provisions, IPPs are prohibited from registering, monetizing, or otherwise utilizing Environmental Attributes outside of the approved frameworks, unless prior consent has been obtained from the Department. The Department retains the sole discretion to determine the use, or monetization of Environmental Attributes in terms of the relevant environmental policies.

- IPP prohibited from registering Environmental Attributes unless prior consent obtained (§5)
- Prior authorization required otherwise may be inconsistent with contracts (§6)
- Verra to review double counting (including contractual obligations) (§7)

Carbon credits are subject to contractual provisions (§4):

Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers.

In addition to the above, the monetization of carbon credits resultant from any VCS project, would fall outside of the registration and issuance process, as this would be a commercial agreement between project proponents and shareholders regarding any revenue sharing. The VVB reviewed current contracts between the PPs regarding carbon credits as part of the validation process.

IPP prohibited from registering Environmental Attributes unless prior consent obtained (§5): Independent

6. Based on our current understanding, unless prior written authorisation has been obtained from the Department, the IPPs do not hold the exclusive rights to register such environmental attributes under voluntary carbon markets, including the VCS. Accordingly, such registrations, if made without the necessary approval, may be inconsistent with the contractual obligations under which these projects were awarded and developed.

7. Given the potential implications of double-counting, the Department requests that Verra reviews the registration and claims associated with the listed projects, considering the contractual limitations and rights under the REIPPPP and RMIPPPP.

legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity, it was deemed that:

- IPPs are not prohibited from registering, monetizing, or otherwise utilizing Environmental Attributes outside of the approved frameworks,
- the Department does not retain any discretion to determine the use, or monetization of environmental attributes, and
- there are no environmental policies that specifically address these issues except for regulations under the Carbon Tax Act which provide for a process in terms of which project proponents (including project proponents of REIPPPP projects) can seek authorisation to list carbon credits as carbon offsets on the Carbon Offset Administration System. proponent.
- the Carbon Tax Act (2019) and associated Carbon Offset Regulations explicitly recognise credits from eligible international standards, including the VCS, for use against domestic carbon tax obligations. This demonstrates that such credits are legally valid for trade and use in South Africa without any requirement for Departmental consent, including REIPPPP projects previously

registered and issued under CDM or VCS. If additional consent were legally required, such credits would not be recognised and available for trade and retirement under this national compliance framework. The fact that they are accepted by National Treasury under the South African Carbon Tax Act (2019) further confirms that registration and monetisation of credits by IPPs is voluntary and lawful.

Prior authorization required otherwise may be inconsistent with contracts (§6):

Independent legal review by the project proponents came to the conclusion that this is a factually incorrect statement because there are no specific contractual provisions relating to carbon credits and/or other claims to environmental attributes under any of the REIPPPP and/or RMI contractual and other documentation to which this comment refers. Given this lack of contractual specificity:

- IPPs do not need prior authorisation from the Department to register projects under the VCS, indeed a number of REIPPPP projects have been registered and operational for some years, including having transacted VCUs into the carbon. Clearly, therefore no prior authorisation is required.

Verra to review double counting (including contractual obligations) (§7):

In accordance with the VCS Standard (§3.23), project proponents need to demonstrate that there is no double

	counting, or double selling of any GHG emission reductions. This was confirmed by cross referencing a number of local and international registries, including Renewable Energy Certificates (RECs). Further to this, during the project development process, each project owned provided a signed declaration that no other environmental benefits are sought offtake agreements or GHG benefits have been allocated outside of the listed Verra project. These documents formed part of the independent validation by the VVB during the audit process. In addition, during the project development process, each project owned provided a signed declaration that no other offtake agreements or GHG benefits have been allocated outside of the listed Verra project.
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2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Anthesis SA team has significant experience in implementing similar project activities, engaging with communities, project management and environmental and social safeguards aligned with the VCS requirements. Members of the project development team have as a collective, in excess of 50 years' experience in the voluntary carbon market. They have been originating, developing and monetizing carbon credits a company for over 2 decades. The Scatec RMI operational project proponent has extensive renewable energy project experience in the South African region and globally, including similar Renewable Power facilities in Brazil, Tunisia, Philippines, Honduras, Malaysia, Romania and Egypt among others. The integrated teams are well-positioned to implement and operate the facility and deliver the associated GHG emission reductions into the local market.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure taken
Natural and human-induced risks to stakeholders' wellbeing	No natural and human induced Risks identified	Measures to prevent natural and human-induced risks were taken and documented in the EIA report.
Risks to stakeholder participation	No risks identified	The stakeholder engagement process was conducted, and comments received from the stakeholders were properly addressed. No evidence of participation barriers to stakeholders identified.
Working conditions	No risks identified related to working conditions.	All workers and their working conditions are subject to the Basic Conditions of Employment Act of South Africa and labour conditions are covered by periodic inspections and compliance to the South African Department of Labour.
Safety of women and girls	No risks related to women and girls were identified.	The project proponent has taken all measures to ensure the safety of women and girls by implementing various policies as per South African labour and anti-discrimination laws.
Safety of minority and marginalized groups, including children	No risks were identified.	The project proponent ensures the safety of minority and marginalized groups, including children in the project as per the South African labour and anti-discrimination laws.
Pollutants (air, noise, discharges to water, generation of waste, and	Increased levels of air and noise pollution are anticipated during the	Noise reduction actions must be implemented, such as use

release of hazardous materials and chemical pesticides and fertilizers)	construction phase due to higher traffic volumes.	of baffle and noise screens on noisy machines. Dust suppression must take place on main access road; and continuous maintenance of gravel roads must be done to reduce air pollution from higher traffic volumes.
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2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁵	Mitigation or preventative measure(s) taken
Discrimination	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Sexual harassment	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Equal pay for equal work	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Gender equity in labor and work	No risks identified	The project proponents have formal human resource and labor

⁵ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		policies as per the Basic Conditions of Employment Act of South Africa.
Forced labor	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Child labor	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.
Human trafficking	No risks identified	The project proponents have formal human resource and labor policies as per the Basic Conditions of Employment Act of South Africa.

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project proponent adheres to international and national human right laws and the Bill of Rights entrenched within the Constitution of South Africa. Risks related to human rights were assessed during the EIA process.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
Rock slab identified within the project site which is likely to be a grave since no other rocks are present in that vicinity.	The EMPr [18] incorporated mitigation measures that must be put in place prior to construction to exclude sites of significant

cultural or historical (heritage) importance with a 5m buffer from the project boundary.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	All property rights are covered by legally binding lease agreements that cover land tenure, and no at-risk communities were involved or affected.

2.3.5 Benefit Sharing

As stated in Section 2.3.4, the project does not impact on property rights as the land is privately owned and has been secured via a lease agreement; therefore, there is no Benefit Sharing Agreement.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	- Disturbance of terrestrial fauna and flora on site due to construction workers and activities. - Ousting of various fauna on account of long-term changes in the surrounding environment. - Mortality of medium and large terrestrial avifauna due to entrapment between perimeter fences.	The EMPr [18] recommends implementation of the following measures to mitigate the risks: - Conduct an Environmental Awareness Training and induction for all construction staff and personnel. - Identify points of excessive faunal activity and impact on operations. Undertake monitoring of faunal activities within the fenced area of the site and the immediate proximity of the site.

		- Facility to be fenced with a single perimeter fence or with two fences that are at least 4 meters apart to avoid entrapment of avifauna.
Soil degradation and soil erosion	- Topsoil loss due to clearance of land surface during construction phase. - Soil erosion by wind or water due to alteration of the land surface characteristics.	The EMPr [18] addressed the risk of soil degradation by implementing the removal of at least 300mm of topsoil prior to construction, which was stockpiled and replaced into disturbed areas on completion of construction. In addition, erosion and sedimentation into water bodies must be minimised through effective stabilisation (gabions and Reno mattresses or similar) and re-vegetation of any disturbed riverbanks.
Water consumption and stress	Potential impact on groundwater availability due to use for cleaning solar panels, and groundwater availability as a result of accidental oil spillages and fuel leakages.	To reduce groundwater consumption, all buildings are to be fitted with rainwater collection and storage systems to supply water to taps and toilets, as well as any outdoor requirements such as cleaning of solar panels. Storage of hazardous/toxic liquids must take place within the contractors' site camp in an area approved by the ECO with a minimum distance of 60 meters away from any freshwater resources.

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes

☐ No

Species and habitat	Plant species that have been identified in the project site are the <i>Aloe dichotoma</i> and the <i>Aloe claviflora</i> which are protected under the provincial Northern Cape Nature Conservation Act. As part of the requirements for an Environmental Authorisation [19] to develop the project in this site, a permit was acquired to continue development with the implementation of a mitigation measure to map the tree species and exclude them from the project boundary.
Areas needed for habitat connectivity	The project site is located in an area that falls within the Bushmanland Arid Grassland which is poorly conserved, despite the presence of drainage lines and wetlands that provide habitats for riparian vegetation and fauna. The EMPr [18] incorporated a sensitivity map to identify all the areas needed for habitat connectivity, which are to be excluded from the project development.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	Loss of protected plant species <i>Aloe dichotoma</i> and <i>Aloe claviflora</i> that are present in the project site.	The EMPr [18] ensures that the project footprint does not include the protected plant species, and the appointed ECO will monitor the presence of the tree species in the project site.
Areas for habitat connectivity	Disruption of ecological processes within the drainage systems and wetlands, as well as loss of habitat for riparian vegetation and fauna.	Project development will exclude all drainage systems and wetlands present in the project site by use of a 32 meter buffer zone surrounding the hydrological features.

2.4.2 Introduction of species

The EMPr [18] contains management interventions and mitigation measures for controlling alien invasive species that may occur on the site. No species have been introduced as a direct part of the project activities.

The following existing alien species are located in the project area.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
<i>Atriplex lindleyi</i>	- Pre-construction clearance of all exotic vegetation on site should be undertaken to reduce the possibility of further exotic weed invasion. - Design and implement an exotic weed control plan for the site for the duration of the operational phase.
<i>Datura ferox</i>	- Pre-construction clearance of all exotic vegetation on site should be undertaken to reduce the possibility of further exotic weed invasion. - Design and implement an exotic weed control plan for the site for the duration of the operational phase.
<i>Prosopis glandulosa</i>	- Pre-construction clearance of all exotic vegetation on site should be undertaken to reduce the possibility of further exotic weed invasion. - Design and implement an exotic weed control plan for the site for the duration of the operational phase.

The table below identifies and summarizes any risks related to invasive species.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	A risk of establishment and spread of alien invasive plants due to the clearing and disturbance of indigenous vegetation exists	An exotic weed control plan was developed as part of the EMPr [18] to prevent introduction of invasive species. Additionally, vehicles used in the development site should be regularly washed to avoid the dispersal of seeds of any exotic or invasive species into the project site.

2.4.3 Ecosystem conversion

The project activity is not in any of the categories; ARR, ALM, WRC or ACoGS.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	N/A	This project is not an ARR, ALM, WRC or ACoGS project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table below includes the methodology(s), tools, and modules applied to the project activity.

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL 01	Tool for the demonstration and assessment of additionality	7.0
Tool	TOOL 07	Tool to calculate the emission factor for an electricity system	7.0
Tool	VT0011	Electricity System Emission Factors	1.0
Tool	TOOL 23	Methodological tool: Additionality of first-of-its-kind project activities	3.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable), as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s);	a) The project activity involves construction and operation of grid-connected solar power plant integrated with BESS, at a site where no renewable energy power plant was operated prior to the implementation of the

Methodology ID	Applicability condition	Justification of compliance
	(c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	project activity and therefore a “green field power project” as per the definition of the methodology and hence complies to the applicability condition (a). Hence the project activity meets the applicability condition of the methodology.
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s); (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is	This project activity is a grid-connected renewable energy power generation project which includes the integration of a BESS into the greenfield solar power plant. So, the project meets applicability condition (a)

Methodology ID	Applicability condition	Justification of compliance
	located at site of the greenfield renewable power plant.	
	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the	(a) This project is a hybrid renewable energy plant utilising solar power with BESS. (b) This project is not a capacity addition, retrofit, rehabilitation or replacement, but a greenfield plant. (c) BESS is an integral part of the project design and is a requirement of the project approvals and power purchase agreement. (d) BESS will only be charged from the greenfield

Methodology ID	Applicability condition	Justification of compliance
	associated renewable energy power plant(s). Only during exigencies² may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	renewable energy (solar) plants at each location. No grid power will be used to charge the BESS component. (e) Not applicable to this project as the project does not employ PSP
	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density,	Not applicable to this project activity as it does not employ hydro power.

Methodology ID	Applicability condition	Justification of compliance
	calculated using equation (7), is greater than 4 W/m²; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or	Not applicable to this project as the project does not employ hydro power

Methodology ID	Applicability condition	Justification of compliance
	(b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	In the case of PSP, the project participants shall demonstrate in the VCS Project Description that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable to this project as the project does not employ PSP
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	(a) Not applicable to this project as the project does not employ any fuel switching. (b) Not applicable to this project as the project does not employ any biomass firing

Methodology ID	Applicability condition	Justification of compliance
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	Not applicable to this project as the project does not employ any retrofits, rehabilitations, replacements, or capacity additions

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 01	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Since the applied methodology is not a new methodology, project proponent has applied this tool for the demonstration additionality in compliance with the tool. Refer to section 3.5 below for the detailed applicability of this tool and additionality assessment. Hence this tool is applicable.
CDM TOOL 01	Once the additionality tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	In line with the methodology requirement PP has applied this tool for the demonstration of additionality assessment. Hence this tool is applicable.

Methodology ID	Applicability condition	Justification of compliance
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CDM TOOL 07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This tool is applied to estimate the OM, BM and/or CM for the project activity
CDM TOOL 07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The emission factor for the project electricity system is calculated for grid power plants only.
CDM TOOL 07	In case of CDM projects the tool is not applicable if the project electricity	The project activity is not a CDM project, however the project activity is located in a

	system is located partially or totally in an Annex I country.	non-Annex I country and hence the tool is applicable.
CDM TOOL 07	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	N/A as the project activity does not involve biofuels.

Methodology ID	Applicability condition	Justification of compliance
TOOL VT 0011	This tool is applied to calculate the CO ₂ emission factor of an electricity system whose dispatch and generation are affected by a project activity. It is globally applicable.	The tool is applicable as the dispatch and generation is affected by a project activity.
TOOL VT 0011	All applicability conditions of the most recent version of TOOL07 must be met, except for Applicability Condition 5, which is removed under this revision	As per the table above, all applicability conditions of the most recent version of TOOL07 are met.
TOOL VT 0011	This tool must be used in conjunction with the most recent version of TOOL07. All the procedures and requirements of TOOL07 must be applied unless this tool indicates otherwise.	The tool has been used in conjunction with CDM TOOL07, as per the applicability conditions above and in section 5.3 the Grid Emission Factor determination.

Methodology ID	Applicability condition	Justification of compliance
CDM TOOL 23	This methodological tool is applicable to project activities that wish to use the "first-of-its kind" approach to demonstrate additionality and that use versions of baseline and monitoring methodologies, or the "Tool for the demonstration and	The project activity applied CDM methodology ACM0002 and Tool 01, which allow using the "first-of-its-kind" approach for demonstrating additionality. Hence this tool

	assessment of additionality" or the "Combined tool to identify the baseline scenario and demonstrate additionality", which allow using the "first-of-its-kind" approach for demonstrating additionality.	is applicable to this project activity.
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3.3 Project Boundary

As per the methodology ACM0002 version 22, the project boundary includes the power plants, and all power plants connected physically to the electricity system that the project power plant is connected to.

The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions.

Source		Gas	Included?	Justification/Explanation
Baseline	CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to project activity.	CO ₂	Yes	Main emission source (Grid electricity)
		CH ₄	No	Excluded for simplification. Minor source of emissions. Conservative
		N ₂ O	No	
		Other	No	No other GHG emissions relevant.
Project	Greenfield solar photovoltaic electricity generation project activity	CO ₂	No	No CO ₂ emissions from project activity. Minimal taken as negligible
		CH ₄	No	Excluded. Minimal taken as negligible
		N ₂ O	No	Excluded. Minimal taken as negligible
		Other	No	No other GHG emissions
	Charging of BESS using electricity from the renewable energy plants	CO ₂	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
		CH ₄	No	Not accounted as BESS will only be charged from the renewable energy

Source	Gas	Included?	Justification/Explanation
			generation plants on site therefore taken as negligible.
	N ₂ O	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.
	Other	No	Not accounted as BESS will only be charged from the renewable energy generation plants on site therefore taken as negligible.

The below map clearly shows the physical locations of the project activity based on the description provided in Section 0 (Description of the Project Activity) above.

The diagram the equipment, systems and flows of mass and energy is also included in figure 2 below.



Figure 6: Physical location of project activity

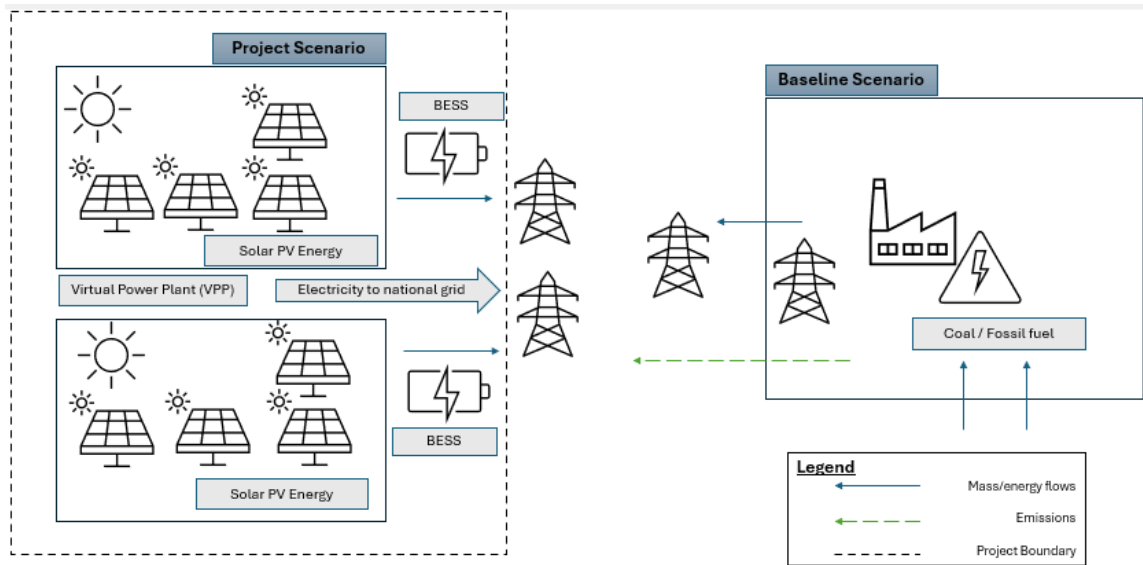


Figure 7: Project boundary, mass & energy flows of Scatec RMI Cluster Project

3.4 Baseline Scenario

The baseline scenario for the Scatec RMI Hybrid Cluster project is defined as the continuation of current grid operations, where electricity demand is primarily met by existing grid-connected fossil fuel (predominantly coal) power plants, and the addition of new fossil fuel-based capacity/generation sources. This approach aligns with the requirements of the ACM0002 (Version 22.0) methodology, which states in Section 5.2.1: “If the project activity is the installation of a Greenfield power plant with or without Battery Energy Storage System (BESS), the baseline scenario is electricity delivered to the grid by the project activity that would otherwise have been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in AM TOOL07 ‘Tool to calculate the emission factor for an electricity system (Version 07.0)’.

Justification of the Baseline Scenario:

The South African electricity grid is characterized by high carbon intensity due to its heavy reliance on coal-fired power plants, which accounts for 80% of its energy mix [20]. This high reliance on coal positions South Africa among the top greenhouse gas emitters per capita globally [21].

According to South Africa’s Integrated Resource Plan (IRP 2019) [22] coal remains a dominant energy source, projected to account for over 50% of the electricity generation mix through 2030. According to South Africa’s Integrated Resource Plan (IRP 2019) [22], coal remains a dominant

energy source, projected to account for over 50% of the electricity generation mix through 2030. The IRP outlines significant reliance on existing coal-fired power plants, with limited expansion of renewable energy capacity due to financial, regulatory, and infrastructural barriers. Despite government efforts to increase renewable energy through initiatives like the Renewable Energy Independent Power Producer Procurement Programme (REIPPPP), which the current projects forms a part of, the expansion pace has been slow, and integration challenges persist, resulting in continued reliance on fossil fuel generation. The South African energy sector faces numerous challenges, including delays in project approvals, difficulties in securing financing, and grid connectivity issues. These barriers significantly impact the deployment of renewable energy projects, validating the assumption that the baseline scenario will continue to be dominated by fossil fuel-based power plants, including both existing and new coal and gas-fired units.

Historical data from Eskom, the primary electricity supplier, indicates a stable grid emission factor of approximately 0.932 tCO₂e/MWh over the past five years, driven by a heavy reliance on coal-fired plants. This trend is expected to continue in the absence of major interventions like the de Aar Hydra project, further supporting the selection of this baseline scenario. The grid emission factor for this baseline is calculated in accordance with the CDM TOOL07 “Tool to calculate the emission factor for an electricity system” Version 07.0 and VT0011 “Electricity System Emission Factors” Version 1.0. Option A under the Simple OM approach was selected along with the default weighting for solar projects in the first crediting period. This results in a combined margin factor for the baseline grid of 0.932 tCO₂e/MWh. The steps taken to calculate the grid emission factor as per CDM TOOL 07 and VT011 are outlined in section 4.1 below. §28 – 34 of ACM0002 do not apply to this project, as shown in Table 4 below:

Table 4: Identification of Baseline scenario as per ACM0002

Section number in ACM0002	Applicable to	Applicability to the project
5.2.2	Capacity addition to an existing renewable energy power plant or integration of a BESS to an existing solar photovoltaic or wind power plant/unit	N/A – the project is a Greenfield facility and not a capacity addition to an existing renewable energy power plant. The BESS components are also greenfield and exclusive to the greenfield De Aar Hydra generation and storage facilities.

5.2.3	Baseline scenario for retrofit or rehabilitation or replacement of an existing power plant	N/A – the project is a Greenfield facility and not a retrofit, rehabilitation or replacement of an existing power plant.
5.2.4	Baseline scenario for retrofit of an existing solar photovoltaic or wind power plant/unit with the integration of BESS	N/A – the project is a Greenfield facility and not a retrofit of an existing facility.
5.2.5	Baseline scenario for pumped storage projects operating in coordination with the renewable energy plants	N/A - the project is a Greenfield facility and is not connected to a Greenfield pumped storage power plant.

The project activity is the installation of a greenfield solar PV power plant with associated BESS components. The baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07 and VT0011.

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

3.5.2 Additionality Methods

Additionality is demonstrated by proving that the Project is first-of-its kind in the geographic area of South Africa. The following tools were used to evaluate additionality:

- CDM TOOL 01 version 7.0 Methodological Tool for the demonstration and assessment of additionality
- CDM TOOL 23 version 3.0 Methodological Tool for assessing additionality of first-of-its-kind project activities

CDM TOOL 01 version 7 prescribes a stepwise approach for evaluating additionality, and if Step 0 is satisfied the project is additional:

Step 0: Demonstrate whether the proposed activity is the first-of-its-kind.

This project activity is considered a first-of-its-kind project, as it is the first large scale project using BESS technology, powered by Solar PV, to be implemented within the country.

CDM TOOL 23 version 3.0 describes the first-of-its-kind approach for demonstrating additionality, and justification for these requirements is shown in the table below:

TOOL 23. Identification of a first-of-its-kind project activity	Justification of compliance
1. The project is the first in the applicable geographical area that applies a technology that is different from technologies that are implemented by any other project, which are able to deliver the same output and have started commercial operation in the applicable geographical area before the project design document [<i>Project Description (PD)</i>] is published for global stakeholder consultation or before the start date of the proposed project activity, whichever is earlier.	i. The applicable geographic area is the entire host country (i.e. Republic of South Africa); ii. The proposed project activity is the first project of this scale (150MW collective contracted capacity) using BESS technology, powered by solar PV technology to reach commercial operation in South Africa Other projects that apply the same combination of technologies are shown in Table 4 and their outputs range from 75MW to 150MW contracted capacity. While the smallest was a factor of 50% lower and none exceeded the project capacity. They were all considered in the comparison iii. Only the Scatec RMI Hybrid Cluster had reached commercial operation at the time of this Validation, as it reached initial COD on 19 November 2023 [11]and achieved completed COD for the entire project capacity on 9 December 2023. This is earlier than any of the other comparable projects.

	Evidence to support the above claims is publicly available, as shown below.
2. Measure - project implements one or more of the following four emission reduction measures: (a) Fuel and feedstock switch (example: switch from naphtha to natural gas for energy generation, or switch from limestone to gypsum in cement clinker production); (b) Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies (example: energy efficiency improvements, power generation based on renewable energy); (c) Methane destruction (example: landfill gas flaring); (d) Methane formation avoidance (example: use of biomass that would have been left to decay in a solid waste disposal site resulting in the formation and emission of methane, for energy generation).	i. The proposed project activity uses renewable energy to charge the BESS, therefore is allowable under Tool 23 2.(b) and battery storage/BESS is not excluded under the VCS Standard version 4.7. ii. This project does not include industrial gases, transport and afforestation/reforestation projects – which are not allowed under Tool 23.
3. Output - goods/services produced by the project activity includes, among other things, heat, steam, electricity, methane, and biogas unless otherwise specified in the applied methodology.	i. Electricity generated from renewable sources like solar PV, is allowable under Tool 23, and battery energy storage systems are not excluded under the VCS Standard version 4.7.
4. Different technologies - technologies deliver the same output and differ by at least one of the following: (a) Energy source/fuel (e.g. energy generation by different energy sources such as wind and hydro and different types of fuels such as biomass and natural gas); (b) Feed stock (e.g. production of fuel ethanol from different feed stocks such as sugar cane and starch, production of cement with varying percentage of	i. The proposed project activity is a large-scale project, with the technology being renewable energy (Solar PV) linked to a BESS with a hybrid power plant controller is to have a combined energy capacity of 495 MWh, with 225 MW dispatch capacity at the Delivery Point and 150 MW contracted capacity. The capacity is >15MW and therefore a large scale project; ii. The BESS with its associated hybrid power plant controller is charged from the solar PV plant, split over three footprints located adjacent to one another and forming a

alternative fuels or less carbon-intensive fuels); (c) Size of installation (power capacity) or energy savings: (i) Micro (<5MW); (ii) Small (<15MW); (iii) Large (>15MW)	combined total installed capacity of an approximately 540 MW peak. iii. A comparison with other projects awarded 'preferred bidder' status under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP) Bid Submission process, conducted from August 2020 until 1 June 2021 is shown in Table 5 (below) and further evidence from the NERSA website (Figure 8). From this List of Additional Preferred Bidders, Scatec Kenhart cluster was the first to reach commercial operation and uses similar BESS with RE technologies as the other projects (with the exception of Mulilo Total Coega, a Solar PV project only) and the three Karpowership projects (proposed as gas fired floating power plant). The RMIPPPP process is the only process in the host country that allows for projects of this scale to supply energy to the national grid and no previous or future rounds have been repeated at the time of this Validation.
5.First-of-its-kind project activities select a crediting period that is "a maximum of 10 years with no option of renewal".	i. A 10 year crediting period was selected with the start and end dates of the 10-year fixed crediting period are 19-November-2023 to 18-November-2033.

Evidence to demonstrate this project activity is the first-of-its-kind for BESS powered by renewable energy, is found on the REIPPP webpage. The South African government, via its Department of Mineral Resources and Energy (DMRE), National Treasury (NT) and the Development Bank of Southern Africa (DBSA) established the IPP Office for the specific purpose of transparent reporting and delivering on the Independent Power Producers Procurement Programme (IPPPP) objectives⁶. The REIPPPP is a competitive bidding process under the Electricity Regulation Act (2006) (ERA) [23] and is used by the South African government to secure electricity from private entities, known as Independent Power Producers (IPPs), who use renewable energy sources. It facilitates private sector investment on a voluntary basis through competitive bids, while supporting socioeconomic development and improving energy security while promoting local ownership. The REIPPPP and RMIPPPP was a created by the South

⁶ <https://www.ipp-projects.co.za/Home/About>

African government to promote the growth of renewable energy projects in the country. The program works with a request for proposals being issued by the IPP Office inviting bids from private companies. A competitive bidding process and bid evaluation follows with successful bidders being awarded a power purchase agreement (PPA). Project owners are then responsible for developing and constructing their power facilities within agreed timeframes and in accordance with local legislation applicable to the construction of such a facility (e.g. Environmental Impact Assessment, Employment etc.). It was introduced in 2011 to lessen the national reliance on fossil fuel energy and encouraged private sector investment in onshore wind, photovoltaic, concentrated solar power (CSP), biomass, landfill gas and small hydro technologies [24]. The REIPPP is a voluntary program for private sector to participate in the national energy mix and is not mandated by any legislation. It is one of the energy options outlined in the National Development Plan and the Integrated Resources Plan of 2010. In bid windows 5 and 6 of the REIPPP, 6000 MW of additional generation capacity, including BESS solutions was advertised under the Risk Mitigation Independent Power Producer Procurement Programme (RMIPPPP). In the evaluation of the first of its kind for this project, all RMIPPPP projects were considered, in addition to any projects that may have a BESS component and are connected to the national grid. This was felt to be conservative and it expanded the evaluation to a greater pool than just BESS technology projects. All RMIPPPP projects that were successfully awarded are listed on the Risk Mitigation Project section of the Independent Power Producers (IPP) Office website⁷, the following projects are listed:

Table 5: List of preferred bidders announced in March and June 2021 by IPP Office

⁷ <https://www.ipp-rm.co.za/>

Project name	Technology	Contract capacity (MW)	Commercial Operation Status	Comments
Scatec RMI Hybrid Cluster (project under consideration)	BESS (Solar PV)	150	19 November 2023	First project using this technology and of this scale to start commercial operation (VCS project # 5488)
Oya Hybrid	BESS (Solar PV & Wind)	128	Planned operation in future (October 2025)	Still under construction (VCS project # 5312)
ACWA Power DAO	BESS (Solar PV)	150	Planned operation in future (Q2 2026 ⁸)	Project signed PPA in December 2023 with planned operation scheduled for Q2 2026. Construction had not commenced at the time of this assessment.
Umoyilanga Energy	BESS (Solar PV & Wind)	75	Planned operation in future (October 2025)	Still under construction (VCS project # 5272)
Karpowership SA Coega	Gas fired floating power plant	450	On hold ⁹	Electricity and Energy Minister Kgosietsso Ramokgopa announced deal was no longer viable in October 2024. Technology does differ from project activity.
Karpowership SA Richards Bay	Gas fired floating power plant	450	On hold ¹⁰	Electricity and Energy Minister Kgosietsso Ramokgopa announced deal was no longer viable in October 2024. Technology does differ from project activity.

Karpowership SA Saldanha	Gas fired floating power plant	450	On hold ¹¹	Electricity and Energy Minister Kgosietsso Ramokgopa announced deal was no longer viable in October 2024. Technology does differ from project activity.
Mulilo Total Coega	Solar PV	198	Construction has not started yet ¹²	Project has been renamed and sold and continue with new ownership ¹³ . Technology is solar PV only and construction had not commenced at the time of assessment.
Mulilo Total Hydra Storage	BESS (Solar PV)	75	Planned operation in future (September 2025)	Project has been renamed and sold and continue with new ownership ¹⁴ . Still under construction (VCS project # 5337).

Further to this, as of August 2024, the National Energy Regulator of South Africa (NERSA), lists all section 34 IPPs that have been awarded contracts under the REIPPP programme. [25] Only 7 BESS projects are listed which includes the 3 portions of the Scatec RMI Hybrid Cluster project. [25] All these projects are all already included in Table 5 above and this therefore provides further corroborating evidence to show the Scatec RMI Hybrid Cluster was the first to reach commercial operation, and is therefore the first-of-its-kind in South Africa, as there were no other projects on the list that were either

⁸ <https://www.acwapower.com/news/acwa-power-signs-power-purchase-agreement-for-south-africas-largest-hybrid-dispatchable-renewable-energy-project/>

⁹ <https://www.engineeringnews.co.za/topic/energy>

¹⁰ <https://www.engineeringnews.co.za/topic/energy>

¹¹ <https://www.engineeringnews.co.za/topic/energy>

¹² <https://www.power-technology.com/data-insights/power-plant-profile-mulilo-total-coega-solar-pv-park-south-africa/>

¹³ <https://www.mulilo.com/news/2023/mulilo-energy-holdings-risk-mitigation-ipp-procurement-programme/>

¹⁴ <https://www.mulilo.com/news/2023/mulilo-energy-holdings-risk-mitigation-ipp-procurement-programme/>

part of the RMIPPP or had a similar technology of renewable energy (in this case solar PV, but wind was also considered) linked to a BESS with a hybrid power plant controller.

110	REIPPP BWS	RE_PV_0037	Grootfontein PV 3	Grootfontein PV 3 (RF) Proprietary Limited	SOLAR PV
111	RMIPPP	RM-TA-0025	ACWA Power Project DAO	ACWA Power Project DAO (RF) Proprietary Limited	Solar PV (BESS)
112	RMIPPP	RM-TA-0028	Umoyilanga Energy	Umoyilanga (RF) Proprietary Limited	Hybrid (BESS)
113	RMIPPP	RM-TA-0180	TotalEnergies Hydra Storage	Total Energies Hydra Storage (RF) Proprietary Limited	Solar PV (BESS)
114	RMIPPP	RM-TA-0032	Scatec Kenhardt 1	Scatec Kenhardt 1 (RF) Proprietary Limited	Solar PV (BESS)
115	RMIPPP	RM-TA-0195	Scatec Kenhardt 2	Scatec Kenhardt 2 (RF) Proprietary Limited	Solar PV (BESS)
116	RMIPPP	RM-TA-0195	Scatec Kenhardt 3	Scatec Kenhardt 3 (RF) Proprietary Limited	Solar PV (BESS)
117	RMIPPP	RM-TA-0006	Oya Energy Hybrid Facility	Oya Energy Proprietary Limited	Solar PV (BESS)
118	DOE PEAKERS	N/A	Avon	Avon Peaking Power (RF) (Pty) Ltd	Open Cycle Gas Turbine

Figure 8: Extract from NERSA Annex 1 list of IPP projects - BESS Technology only¹⁵

Further to this, a project search was conducted (including during the development of a national Grid Emission Factor) using TOOL 07 which considers CDM, VCS and Gold Standard plants in the Build Margin Calculation (Table). Not one of the 26 projects from the resultant search including those with renewable energy source (either solar PV or wind or a hybrid combination) linked to BESS with a hybrid power plant controller had been constructed or started operation prior to the current project under consideration. This confirmed that none of these plants applied a similar technology to the current project for dispatchable power so were not considered in the determination of this step of additionality and gave further evidence of the first of its kind nature of the current project with its solar PV renewable energy source (but also considering solar PV or wind or a hybrid combination) linked to BESS with a hybrid power plant controller as the Scatec RMI Hybrid Cluster project had completed construction and feed to the grid prior to any of the other projects in the country.

Further to this in the presence of the then Minister of Mineral Resources and Energy, Mr. Gwede Mantashe; Acting Director General of the Department of Mineral Resources and Energy, Ms. Patricia Gamede; Eskom CEO Mr. Andre de Ruyter, Head of the IPP Office, Mr. Bernard Tshifhiwa Magoro stated that the ,*“Scatec projects signed on 02 June 2022, are a “first of its kind” in South Africa, combining Solar PV and Battery technologies to contribute 150MW of dispatchable capacity to the national grid.”*[26] Supplemental to this the Department of Mineral Resources and Energy announced in July 2022 that, *“Once completed, the (..) power stations will be the first dispatchable renewable energy projects in South Africa.”* [2][More recently it was confirmed that the hybrid project was, *“one of the world’s first and largest hybrid solar and battery storage facilities.”* [27]

The conclusion of this step is that the project can be considered the first project of its kind (renewable energy source (either solar PV or wind or a hybrid combination) linked to BESS with a hybrid power plant

¹⁵ Note that this NERSA table erroneously indicates Oya Energy Hybrid Facility as a Solar PV (BESS) project, but it is actual fact a Hybrid BESS similar to Umoyilanga Energy facility.

controller) in South Africa to reach its Commercial Operation Date (COD) and start feeding electricity into the national grid, as outlined in the Project Start Date above. It has demonstrated that it is a first-of-its-kind activity and as per §13 of TOOL 23 and §4.1.18 of TOOL 01 therefore the project is considered additional and it is not required to proceed to steps 2, 3 or 4 as per Figure 9 below which is extracted from CDM TOOL 01.

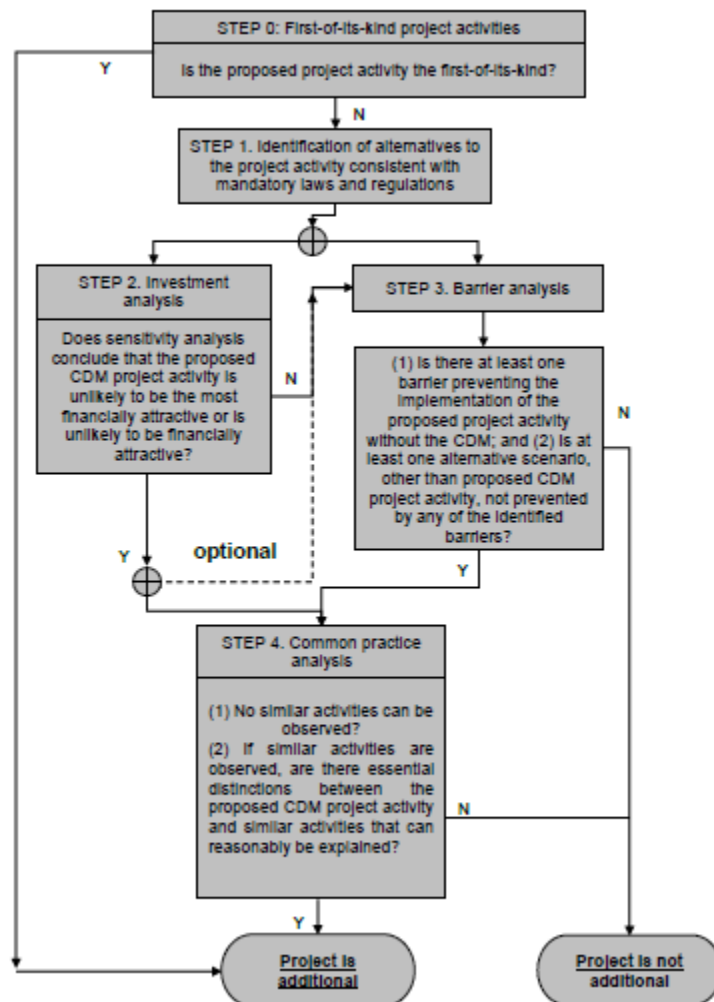


Figure 9: Flowchart of the step-wise additionality approach from CDM TOOL 01

3.6 Methodology Deviations

There are no methodology deviations applied in the Project.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The Project achieved its Commercial Operation Date (COD) with the signature of the Notice of Commencement of the Facility between the facility operating entity and the national power utility formally commencing the Power Purchase Agreement. This was achieved for the first portion of the project on 19 November 2023 [11]. This was the start date of the project and GHG reduction activity.

From this date, the plant operated and the second portion of the project achieved COD on 26 November 2023 [28] while the remaining portion on 9 December 2023 [29]. No other changes from the project implementation outlined in this document were experienced in this initial verification period. There were also no changes to the project proponents.

While the plant experienced normal operations during the monitoring period, the following abnormal incidents were recorded and managed during the period. There were no incidents in the current monitoring period that impacted GHG emission reduction and monitoring.

Incident	Impact
Grid event – Eskom overhead line & current transformer (CT) failure.	Reduction of a portion of plant supply while incident was repaired
Setpoint instability after initial grid events.	OEM intervention and faults rectified with preventive actions. Limited production loss.
Overhead line pole failure.	Limited production loss while incident was repaired.
Power Plant Controller (PPC) set point instability.	OEM intervention and faults rectified with preventive actions. Minimal production loss.
Lightning strikes on overhead lines.	Continued risk in thunder storms. Each incident resulted in limited production loss while incident was repaired

BESS software issues.	OEM intervention and faults rectified with preventive actions and local capacity knowledge transfer. Limited production loss.
Power conversion system (PCS) overheating.	Root cause analysis intervention and faults rectified with preventive actions including planned maintenance. Limited production loss.

Chronology of important dates:

Activity	Date
Bid Submission	Dec 2020
PB Award	June 2021
Financial Close	July 2022
Commercial Operation	SK1 – 19 Nov 2023 SK2 – 26 Nov 2023 SK3 – 09 Dec 2023
Signing Contract with VVB	24 March 2025
Listing project for GSC	21 April 2025 (GSC start date)

5 Quantification of Estimated GHG Emission Reductions and Removals

5.1 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as per the calculation steps described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Equation 1 (ACM0002 Equation 11)

Where:

BE_y	=	Baseline emissions in year y (tCO ₂ /yr)
$EG_{PJ,y}$	=	Net electricity fed into the grid supplied by the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO ₂ /MWh)

The results obtained are: 810 840 t CO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	810, 840	t CO ₂ / yr
$EG_{PJ,y}$	870, 000	MWh/yr
$EF_{grid,CM,y}$	0.932	t CO ₂ /MWh

The plant's auxiliary electricity consumption is supplied by its connection to South Africa's national grid (Eskom).

$EG_{PJ,y}$ represents the net electricity supplied to the grid, which is measured as the difference between electricity exported to the grid and electricity imported from the grid. The plant does not rely on the on-site diesel generators for electricity generation, it only serves as emergency backup. Since the plant does not rely on the backup diesel generators to meet its auxiliary electricity supply needs, the emissions from such generator are considered negligible as per paragraph 42 of ACM0002. Fossil fuel usage will, however, still be monitored for record-keeping purposes.

The project is a greenfield power plant and therefore $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{facility,y}$$

Equation 2 (ACM0002 Equation 12)

Where:

$EG_{PJ,y}$ = Net electricity generation fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Net electricity generation supplied by the project plant to the grid in year y (MWh/yr)

The results obtained are: 870,000 MWh / yr (from technical specifications)

Grid Emission Factor Calculation

The grid emission factor for South Africa's national electricity grid was calculated in accordance with the following tools:

- CDM Tool 07 version 7.0 [30]
- VCS Tool VT0011 version 1.0 [31]

According to these tools, the combined margin (CM) grid emission factor is calculated as the weighted average of the system's operating margin (OM) and build margin (BM). The operating margin refers to the group of existing power plants whose current electricity generation would be affected by the proposed renewable energy project activity. The build margin represents the likely future power plants in the electricity systems whose construction and future operation would be affected by the proposed renewable energy project activity. This calculation is only applicable to grid-connected power plants and does not include off-grid power plants.

In accordance with Tool07 The following 6 steps are followed:

- **Step 1:** Identify the relevant electricity systems

- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional)
- **Step 3:** Select a method to determine the operating margin (OM)
- **Step 4:** Calculate the operating margin emission factor according to the selected method
- **Step 5:** Calculate the build margin (BM) emission factor
- **Step 6:** Calculate the combined margin (CM) emission factor.

5.1.1 Step 1: Identify the project electricity system

The project electricity system was identified, using Option 2, as: The national grid of South Africa – administered by the national power utility, Eskom Holding SOC Ltd, including generation from Independent Power Producers (IPPs) and imports from the South African Power Pool (SAPP) network. This is shown schematically in Figure 10 below.

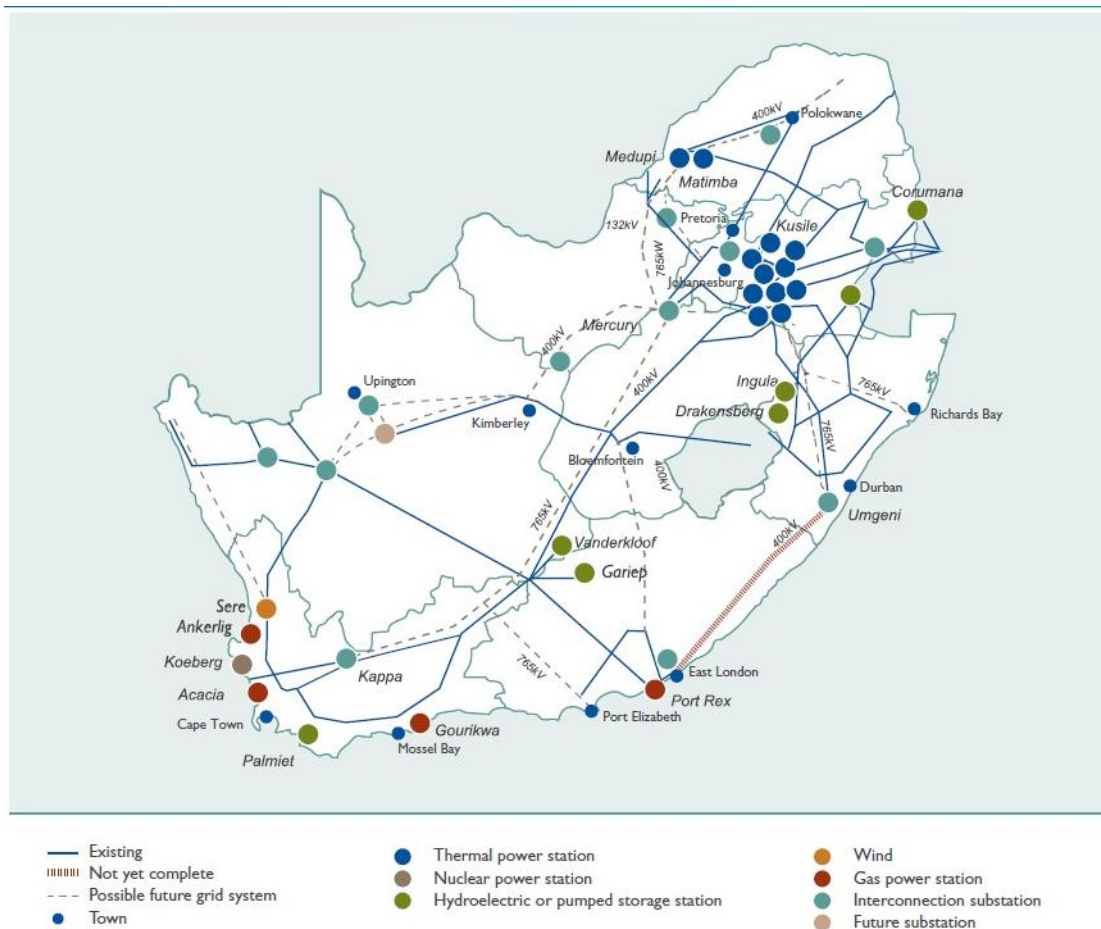


Figure 10: Project electricity system Source: [20]

South Africa's electricity grid is not connected to an electricity system located partially or totally in an Annex I country.

5.1.2 Step 2: Choose whether to include off-grid power plants in the project electricity system

Only grid connected plants (shown in **Table 8**). were included in this calculation.

5.1.3 Step 3: Operating Margin method selection

The operating margin (OM) emission factor ($EF_{grid,OM,y}$) was calculated based on the simple OM method (paragraph 38.a in TOOL07). This was decided based on the following criteria:

- The share of electricity supplied by Low-Cost/Must-Run (LCMR) plants in the last 5 years has ranged from 19% – 23%. It has been consistently lower than 50%, as calculated using TOOL07 40(a) Approach 1, by **Equation 3** below:

$$Share_{LCMR} = average \left[\frac{EG_{LCMR(y-4)}}{total_{y-4}}, \dots, \frac{EG_{LCMR(y)}}{total_y} \right]$$

Equation 3 (Tool 07, 40 Equation 1)

Where:

$Share_{LCMR}$	=	Share of the low cost/must run resources (%)
EG_{LCMRy}	=	Electricity generation supplied by LCMR plants in year y (MWh)
$total_y$	=	Total electricity supplied to the project electricity system in year y (MWh)
Y	=	The most recent year for which data is available

The average share of LCMR plants ($share_{LCMR}$) was found to be equal to 21.2% over the last 5 years.

As per paragraph 45 of VT0011, power plants registered under the VCS program or any other GHG program are included in the sample group used to calculate the operating margin, not including wheeling agreements. As per paragraph 47(a) of tool07, option A was chosen based on the fact that net electricity generation and fuel usage data is available for each power plant serving the system.

The ex-ante option (TOOL07 paragraph 42(a)) was chosen to determine the simple OM. The 3-year generation-weighted average emission factor was calculated based on the most recent data available at the time of listing the project on Verra's registry. For this project, the most recent data available is 2024. Therefore, the generation weighted average of 2022, 2023, and 2024 was used.

The results obtained are shown in **Table 6** below

Table 6: LCMR Analysis

Parameter name	Value	Unit
Share _{LCMR} : 2024	23.6%	Per cent
Share _{LCMR} : 2023	23.1%	Per cent
Share _{LCMR} : 2022	21.3%	Per cent
Share _{LCMR} : Average (5 years)	21.2 %	Per cent
EG _{LCMRy} : 2024	49 751	GWh / yr
EG _{LCMRy} : 2023	49 691	GWh / yr
EG _{LCMRy} : 2022	48 092	GWh / yr
total _y : 2024	210 648	GWh / yr
total _y : 2023	215 318	GWh / yr
total _y : 2022	226 226	GWh / yr

5.1.4 Step 4: Operating margin calculation

The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including LCMR power plants. The following plants were identified as LCMR, and therefore excluded from the OM calculation:

Table 7: LCMR plants serving the project electricity system

LCMR Plant name	Technology	Reason for classification as LCMR
- Gariep - Vanderkloof - Mbashe - First Falls - Ncora - Second Falls	Hydroelectric power stations	Renewable (low generation costs)
- Drakensberg - Ingula - Palmiet	Pumped storage stations	Only used during periods of peak demand (must run)
Sere	Wind energy stations	Renewable (low generation costs)
Koeberg	Nuclear power stations	Nuclear listed as LCMR
- Biomass - Concentrated solar power	Imports and REIPPP plants	Imports or renewable

• Gas/liquid fuel • Hydroelectric • Landfill • Solar PV • Wind		
--	--	--

The power plants used in the OM calculation are shown in **Table 8** below. The simple OM was calculated based on the net electricity generation and the CO₂ emission factor of each power unit (TOOL07 paragraph 47, option A), as per **Equation 4**.

$$EF_{grid,OMsimple,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 4 (Tool 07 48, Equation 3)

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (t CO ₂ /MWh)
$EG_{m,y}$	=	Net electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
y	=	The relevant year as per the data vintage chosen

The results obtained are: $EF_{grid,OMsimple,y} = 1.099 \text{ tCO}_2/\text{MWh}$

All values used are shown in **Table 8**.

Table 8: Operating Margin Plant Data

Plant Name	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Electricity generated (MWh/yr) 2024	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2024 (tCO ₂ /MWh)
Arnot	9 085 064	7 530 158	8 353 868	1.05	1.05	1.07
Duvha	9 822 489	9 555 266	11 787 013	1.07	1.05	1.07
Hendrina	4 738 179	2 136 620	2 305 812	1.16	1.28	1.13
Kendal	15 139 746	15 750 630	12 464 364	1.13	1.10	1.15
Kriel	12 597 519	10 727 877	7 823 567	1.14	1.17	1.22
Lethabo	22 890 481	21 367 191	21 280 064	1.29	1.33	1.37
Matimba	23 180 336	23 257 630	18 926 973	1.05	1.03	1.03
Majuba	19 823 301	15 540 628	18 711 767	1.11	1.18	1.12
Matla	18 412 664	15 170 449	16 707 442	1.07	1.14	1.10
Tutuka	8 524 715	7 564 337	6 394 619	1.21	1.16	1.13
Camden	6 670 341	7 378 745	7 329 029	1.09	1.04	1.11
Grootvlei	2 445 162	2 416 293	1 638 696	1.19	1.22	1.20
Komati	608 728	369 330	0	1.01	0.98	0.00
Kusile	7 992 189	9 119 476	10 262 515	1.10	1.01	0.98
Medupi	22 637 675	23 246 212	22 626 444	0.91	0.87	0.89
Acacia	873	1 069	2 406	1.23	1.18	1.19
Port Rex	937	3 010	4 340	1.51	1.41	1.30
Ankerlig	953 788	1 332 491	1 588 583	0.96	0.96	0.98
Gourikwa	870 396	1 681 217	2 038 316	1.20	1.19	1.18
Total	186 394 583	174 148 629	170 245 818			

The emission factor of each power unit ($EF_{EL,m,y}$) was determined following option A1 (paragraph 49.a) as shown in **Equation 5** below:

$$EF_{EL,m,y} = \frac{\sum_i FC_{i,m,y} \times NCV_{i,y} \times EF_{CO2,i,y}}{\sum EG_{m,y}}$$

Equation 5 (Tool 07 49.(a) Equation 4)

Where:

$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (t CO ₂ /MWh)
$FC_{i,m,y}$	=	Amount of fuel type i consumed by power unit m in year y (Mass or volume unit)
$NCV_{i,y}$	=	Net calorific value of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO ₂ emission factor of fuel type i in year y (t CO ₂ /GJ)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
m	=	All power units serving the grid in year y except low-cost/must-run power units
i	=	All fuel types combusted in power unit m in year y
y	=	The relevant year as per the data vintage chosen

The emission factor for each power unit ($EF_{EL,m,y}$) is shown in Table 8 above.

The values used in Equation 5 are shown in Table 9 below.

$EG_{m,y}$ was obtained from Eskom's website as well as their annual Integrated Reports. [32] [33].

Table 9: Parameters for plant emission factor calculation

Plant Name	Coal usage (t) 2022	Coal usage (t) 2023	Coal usage (t) 2024	Fuel oil usage (t) 2022	Fuel oil usage (t) 2023	Fuel oil usage (t) 2024	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Emission factor coal (tCO ₂ /GJ)	Emission factor fuel oil (tCO ₂ /GJ)
Arnot	5 186 766	4 293 316	4 828 094	39 812	41 497	46 635	19.29	40.4	0.09	0.08
Duvha	5 732 195	5 428 210	6 802 842	35 245	48 518	72 479	19.29	40.4	0.09	0.08
Hendrina	3 024 981	1 486 626	1 426 720	11 013	12 319	9 262	19.29	40.4	0.09	0.08
Kendal	9 362 475	9 395 998	7 712 146	34 004	73 937	113 212	19.29	40.4	0.09	0.08
Kriel	7 838 445	6 865 127	5 183 313	47 351	47 431	56 540	19.29	40.4	0.09	0.08
Lethabo	16 277 939	15 646 686	16 048 047	10 312	13 608	13 252	19.29	40.4	0.09	0.08
Matimba	13 430 751	13 179 578	10 674 314	13 185	9 929	18 175	19.29	40.4	0.09	0.08
Majuba	11 924 693	9 831 151	11 304 146	107 591	153 361	119 808	19.29	40.4	0.09	0.08
Matla	10 842 908	9 536 046	10 147 328	10 737	22 639	19 578	19.29	40.4	0.09	0.08
Tutuka	5 510 214	4 697 013	3 879 070	98 619	73 522	62 137	19.29	40.4	0.09	0.08
Camden	3 932 224	4 129 344	4 402 710	50 673	51 602	56 731	19.29	40.4	0.09	0.08
Grootvlei	1 544 269	1 518 602	1 064 085	38 483	59 904	12 594	19.29	40.4	0.09	0.08
Komati	335 833	197 954	-	1 700	800	-	19.29	40.4	0.09	0.08
Kusile	4 783 403	5 035 930	5 481 821	39 817	16 870	26 640	19.29	40.4	0.09	0.08
Medupi	11 351 788	11 181 866	11 065 548	35 442	18 065	20 489	19.29	40.4	0.09	0.08

Plant Name	Diesel usage (l) 2022	Diesel usage (l) 2023	Diesel usage (l) 2024	Net calorific value Diesel (MJ/l)	Emission factor Diesel (tCO ₂ /GJ)
Acacia	350 693	414 371	936 177	43	0.074
Port Rex	364 144	1 093 406	1 458 236	43	0.074
Ankerlig	300 686 482	418 610 227	507 827 227	43	0.074
Gourikwa	269 555 564	515 526 621	622 893 869	43	0.074

5.1.5 Step 5: Calculate the build margin (BM) emission factor

An ex-ante approach (option 1 in TOOL07 paragraph 72(a)) is selected to calculate the BM emission factor. The most recent information available on units already built for sample group m at the time of submitting the project for validation was used. The procedure followed to determine the sample group of power units (m) used to calculate the build margin is described below. The total electricity generation (AEG_{total}) of all power plants directly supplying the system in 2024 was 210 648 GWh.

In line with VT0011, the set of five power units, including power units registered as VCS or other GHG programs, that started to supply electricity to the grid most recently was identified. Their annual generation was determined ($AEG_{SET-5 \text{ units}}$) as 1 643 833 MWh in 2024, which was below 20% of AEG_{total} .

Table 10: Five most Recent Plants supplying the grid

No.	Plant name	Commissioning Date	Electricity supplied UOM	2024
1	TWE Golden Valley Wind Power Project	2021/05/01	MWh	509 303
2	GS11949 Small-scale solar electrical programme, South Africa	2020/12/16	MWh	22 500
3	Solar Energy Programme for South Africa	2019/02/01	MWh	381 930
4	Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31	MWh	290 500
5	Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31	MWh	439 600
TOTAL				1 643 833

Next, the set of power units that started to supply power to the grid most recently and that comprise 20% of the total annual electricity generation of the electricity system (AEG_{total}) was identified ($SET_{>20 \text{ percent}}$). This set is shown in Table 10 and included units from VCS or other GHG programs. The set's annual generation was determined ($AEG_{set>20 \text{ percent}}$). This was 42 739 622 MWh in 2024.

$AEG_{set>20 \text{ percent}}$ was found to be greater than $AEG_{set 5 \text{ units}}$ and therefore $SET_{>20 \text{ percent}}$ was chosen as SET_{sample} .

The resulting SET_{sample} group of power units, m, used to calculate the build margin is shown **Table**.

Table 11: Plant set for BM Calculation

No.	Plant name	Commissioning Date	Power Produced in Year: 2024 (MWh)
Eskom Fleet Plants			
1	Gourikwa	2007/07/01	2 038 316
2	Medupi	2015/08/22	22 626 444
3	Kusile	2017/08/30	10 262 515
IPP Power Plants			
4	TWE Golden Valley	2021/05/01	509 303
5	GS11949 small scale solar electrical programme, South Africa	2020/12/16	22 500
6	Solar energy programme for south africa	2019/02/01	381 930
7	Longyuan Mulilo de Aar Maanhaarberg Wind Energy Facility	2017/10/31	290 500
8	Longyuan Mulilo de Aar 2 North Wind Energy Facility	2017/10/31	439 600
9	Green Power for South Africa	20/09/2011	1220
10	Grid Connected Wind Power Plant in Witberg	2012/02/01	507
11	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	375
12	South Africa Wind Energy	01/01/2016	95
13	West Coast 1 Wind Farm	01/06/2015	293
14	Renewable Energy Carbon Program for Africa	01/07/2013	140
15	Grid Connected Photovoltaic Renewable electricity generating facilities POA	19/07/2013	21
16	South African Large scale Grid connected Solar Park Programme	05/04/2012	66
17	Cookhouse wind farm in South Africa	01/12/2011	1057
18	Red Cap Kouga Wind Farm	2011/11/02	290
19	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2012/12/01	878
29	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/04/01	95
21	Bokpoort CSP (Concentrating Solar Power) Project, South Africa	2013/01/10	226
22	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	130
23	Electricawinds 30 MW Wind Project at Riverbank	2013/12/01	80
24	Caledon Wind Farm in South Africa	2012/06/01	406
25	Langa Energy Photovoltaic Solar Energy Facility	2012/03/05	140
26	Indwe Wind Project	2012/06/01	148
	TOTAL		42 739 622

As the annual electricity generation of this set comprises 20.29% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample-CDM} > 0.2 \times AEG_{total}$), this sample group is used to calculate the build margin.

The build margin emission factor was calculated as the generation-weighted average emission factor (tCO₂/MWh) for power units m during 2024, which is the most recent year y for which electricity generation data is available, using **Equation 6**:

$$EF_{grid,BM,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Equation 6 (Tool 07 77 Equation 15)

Where:

$EF_{grid,BM,y}$	=	Build margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$EG_{m,y}$	=	Net quantity of electricity delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO ₂ emission factor of power unit m in year y (tCO ₂ /MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

The results obtained are: $EF_{grid,BM,y} = 0.764$

$EG_{m,y}$ and $EF_{EL,m,y}$ were used as shown in **Table 8**.

5.1.6 Step 6: Calculate the combined margin (CM) emission factor

The combined margin (CM) emission factor ($EF_{grid,CM,y}$) was calculated as the weighted average CM, as per **Equation 7**.

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 7 (Tool 07 85 Equation 16)

As per VCSVT011, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM} :

- $w_{OM} = 0.5$
- $w_{BM} = 0.5$

The results obtained are: $EF_{grid,CM,y} = 0.932$ tCO₂/MWh

5.2 Project Emissions

As per the approved consolidated Methodology ACM0002, “Grid-connected electricity generation from renewable sources” (Version 22.0) para 40: “For most renewable energy power

generation project activities, $PE_y = 0$. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

Where,

PE_y = Project emissions in year y (tCO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (tCO₂e/yr)

$PE_{GP,y}$ = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO₂e/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

$PE_{BESS,y}$ = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (tCO₂e/yr)

$PE_{PSP,y}$ = Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (tCO₂e/yr)

As the project activity is the installation of a new grid-connected solar Power plant and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. The Battery Energy Storage System (BESS) was charged exclusively from the project's Solar PV generation under normal operating conditions. No charging of the BESS occurred from the grid or from any fossil fuel-based electricity generators during the monitoring period. On-site fossil fuel generators are for emergency auxiliary backup purposes only. Therefore $PE_{FF,y}$, $PE_{GP,y}$, $PE_{HP,y}$, $PE_{BESS,y}$, $PE_{PSP,y}$ are equal to zero and thus, $PE_y = 0$.

5.3 Leakage Emissions

As per the applied methodology, no other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g.: extraction, processing, transport etc.) are neglected.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per the paragraph 72 of the methodology ACM0002 "Grid-connected electricity generation from renewable sources" Version 22.0 emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y \text{ (Equation (17))}$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emissions in year y (tCO₂e/yr)

PE_y = Project emissions in year y (tCO₂e/yr)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
19-NOV-2023 to 31-Dec-2023	87 841	0	0	87 841	0	87 841
01-Jan-2024 to 31-Dec-2024	810 840	0	0	810 840	0	810 840
01-Jan-2025 to 31-Dec-2025	810 840	0	0	810 840	0	810 840
01-Jan-2026 to 31-Dec-2026	810 840	0	0	810 840	0	810 840
01-Jan-2027 to 31-Dec-2027	810 840	0	0	810 840	0	810 840
01-Jan-2028 to 31-Dec-2028	810 840	0	0	810 840	0	810 840
01-Jan-2029 to 31-Dec-2029	810 840	0	0	810 840	0	810 840

01-Jan-2030 to 31-Dec-2030	810 840	0	0	810 840	0	810 840
01-Jan-2031 to 31-Dec-2031	810 840	0	0	810 840	0	810 840
01-Jan-2032 to 31-Dec-2032	810 840	0	0	810 840	0	810 840
01-Jan-2033 to 18-Nov-2033	722 999			722 999		722 999
Total	8 108 400	0	0	8 108 400	0	8 108 400

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 6, as per TOOL07 “Tool to calculate the emission factor for an electricity system”
Value applied:	0.764
Justification of choice of data or description of measurement methods and procedures applied	The BM was calculated as per option 1 (paragraph 72) of TOOL07 “Tool to calculate the emission factor for an electricity system”. An ex-ante approach was followed, based on the most recent information available on units already built. The BM will be updated for the second crediting period based on the most recent information available at that time. Capacity additions from retrofits of power plants are not included in the calculation of the build margin emission factor.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Operating margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 1, as per TOOL07 “Tool to calculate the emission factor for an electricity system” Data Sources: Atmospheric Emission License (EAL) reports – Eskom Data Portal

Value applied:	1.099
Justification of choice of data or description of measurement methods and procedures applied	The simple OM emission factor was calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. This means that only Eskom's coal-based power plants were included in the calculation. The OM was calculated as per option A (paragraph 47.a) of TOOL07 "Tool to calculate the emission factor for an electricity system", based on the net electricity generation and a CO₂ emission factor of each power unit.
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Source of data	Calculated using Equation 7, as per TOOL07 "Tool to calculate the emission factor for an electricity system"
Value applied:	0.932
Justification of choice of data or description of measurement methods and procedures applied	The combined margin (CM) emission factor (EF_{grid,CM,y}) was calculated based on the weighted average CM method (paragraph 81.a) since the data to determine the BM is available. As per VCSVT011, the following weighted values (for solar power generation) were used for w_{OM} and w_{BM}: • $w_{OM} = 0.5$ • $w_{BM} = 0.5$

	, for the first crediting period and for subsequent crediting periods
Purpose of data	Calculation of baseline emissions
Comments	No additional comments

Data / Parameter	Share _{LCMR}		
Data unit	Percent (%)		
Description	Share of the low cost/must run resources		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website [33] for each power station.[20] and records published on Eskom's website [33] for each power station.		
Value applied	Annual LCMR (%)		
	2022	2023	2024
	21.3%	23.1%	23.6%
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in the Table above		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.		

Data / Parameter	EG _{LCMR,y}
Data unit	MWh
Description	Electricity generation supplied to the project electricity system by the low cost/must run resources in year y

Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2024 [32] and records published on Eskom’s website [33] for each power station.		
Value applied	EG_{LCMRy} : 2024	49 751 000	
	EG_{LCMRy} : 2023	49 691 000	
	EG_{LCMRy} : 2022	48 092 000	
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in the Table above		
Purpose of data	To calculate baseline emissions, using Equation 3.		
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.		

Data / Parameter	total _y		
Data unit	MWh		
Description	Total electricity generation supplied to the project electricity system in year y		
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website [10] for each power station.		
Value applied	<i>total_y</i> : 2024	210 648	GWh/yr
	<i>total_y</i> : 2023	215 318	GWh/yr
	<i>total_y</i> : 2022	226 226	GWh/yr
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in the Table above		
Purpose of data	To calculate baseline emissions, using Equation 3.		

Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.
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Data / Parameter	Y
Data unit	Year
Description	The most recent year for which data is available
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website for each power.
Value applied	2024
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in the Table above
Purpose of data	To calculate baseline emissions, using Equation 3.
Comments	The simple operating margin (OM) can be used where low-cost/must-run resources constitute less than 50 per cent of total grid generation (excluding electricity generated by off-grid power plants) in the average of the five most recent years, determined by using Equation 1 under Approach 1 from section 40 in TOOL 07.

Data / Parameter	$EG_{m,y}$
Data unit	MWh
Description	Net electricity delivered to the grid by power unit m in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website [14] for each power station.
Value applied	

As per **Table 88**

Plant Name	Electricity generated (MWh/yr) 2022	Electricity generated (MWh/yr) 2023	Electricity generated (MWh/yr) 2024
Arnot	9 085 064	7 530 158	8 353 868
Duvha	9 822 489	9 555 266	11 787 013
Hendrina	4 738 179	2 136 620	2 305 812
Kendal	15 139 746	15 750 630	12 464 364
Kriel	12 597 519	10 727 877	7 823 567
Lethabo	22 890 481	21 367 191	21 280 064
Matimba	23 180 336	23 257 630	18 926 973
Majuba	19 823 301	15 540 628	18 711 767
Matla	18 412 664	15 170 449	16 707 442
Tutuka	8 524 715	7 564 337	6 394 619
Camden	6 670 341	7 378 745	7 329 029
Grootvlei	2 445 162	2 416 293	1 638 696
Komati	608 728	369 330	0
Kusile	7 992 189	9 119 476	10 262 515
Medupi	22 637 675	23 246 212	22 626 444
Acacia	873	1 069	2 406
Port Rex	937	3 010	4 340
Ankerlig	953 788	1 332 491	1 588 583
Gourikwa	870 396	1 681 217	2 038 316
Total	186 394 583	174 148 629	170 245 818

Justification of choice of data or description of measurement methods and procedures applied

Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 8

Purpose of data

To calculate baseline emissions, using Equation 4.

Comments

The generation-weighted average CO₂ emissions per unit net electricity generation (t CO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants is used to determine the simple OM emission factor, using Equation 3 under Option A from section 48 in TOOL 07.

Data / Parameter

$EF_{EL,m,y}$

Data unit

tCO₂/MWh

Description	CO ₂ emission factor of power unit m in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.

Value applied	As per Table 8			
	Plant Name	Emission factor 2022 (tCO ₂ /MWh)	Emission factor 2023 (tCO ₂ /MWh)	Emission factor 2024 (tCO ₂ /MWh)
	Arnot	1.05	1.05	1.07
	Duvha	1.07	1.05	1.07
	Hendrina	1.16	1.28	1.13
	Kendal	1.13	1.10	1.15
	Kriel	1.14	1.17	1.22
	Lethabo	1.29	1.33	1.37
	Matimba	1.05	1.03	1.03
	Majuba	1.11	1.18	1.12
	Matla	1.07	1.14	1.10
	Tutuka	1.21	1.16	1.13
	Camden	1.09	1.04	1.11
	Grootvlei	1.19	1.22	1.20
	Komati	1.01	0.98	0.00
	Kusile	1.10	1.01	0.98
	Medupi	0.91	0.87	0.89
	Acacia	1.23	1.18	1.19
	Port Rex	1.51	1.41	1.30
	Ankerlig	0.96	0.96	0.98
	Gourikwa	1.20	1.19	1.18
Justification of choice of data or description of	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power			

measurement methods and procedures applied	Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table above
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.

Data / Parameter	$FC_{i,m,y}$
Data unit	Tonnes
Description	Amount of fuel type i consumed by power unit m in year y
Source of data	Determined from electricity generation data reported in Eskom’s annual Integrated Report for 2024 [32] and monthly records published on Eskom’s website [14] for each power station.
Value applied	As per Table 9

Plant Name	Coal usage (t) 2022	Coal usage (t) 2023	Coal usage (t) 2024	Fuel oil usage (t) 2022	Fuel oil usage (t) 2023	Fuel oil usage (t) 2024
Arnot	5 186 766	4 293 316	4 828 094	39 812	41 497	46 635
Duvha	5 732 195	5 428 210	6 802 842	35 245	48 518	72 479
Hendrina	3 024 981	1 486 626	1 426 720	11 013	12 319	9 262
Kendal	9 362 475	9 395 998	7 712 146	34 004	73 937	113 212
Kriel	7 838 445	6 865 127	5 183 313	47 351	47 431	56 540
Lethabo	16 277 939	15 646 686	16 048 047	10 312	13 608	13 252
Matimba	13 430 751	13 179 578	10 674 314	13 185	9 929	18 175
Majuba	11 924 693	9 831 151	11 304 146	107 591	153 361	119 808
Matla	10 842 908	9 536 046	10 147 328	10 737	22 639	19 578
Tutuka	5 510 214	4 697 013	3 879 070	98 619	73 522	62 137
Camden	3 932 224	4 129 344	4 402 710	50 673	51 602	56 731
Grootvlei	1 544 269	1 518 602	1 064 085	38 483	59 904	12 594
Komati	335 833	197 954	-	1 700	800	-
Kusile	4 783 403	5 035 930	5 481 821	39 817	16 870	26 640
Medupi	11 351 788	11 181 866	11 065 548	35 442	18 065	20 489

Plant Name	Diesel usage (l) 2022	Diesel usage (l) 2023	Diesel usage (l) 2024
Acacia	350 693	414 371	936 177
Port Rex	364 144	1 093 406	1 458 236
Ankerlig	300 686 482	418 610 227	507 827 227
Gourikwa	269 555 564	515 526 621	622 893 869

Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The power plants m are shown in Table 9 Table above
Purpose of data	To calculate baseline emissions, using Equation 5.
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.

Data / Parameter	$NCV_{i,y}$																																											
Data unit	GJ/tonne																																											
Description	Net calorific value of fuel type i in year y																																											
Source of data	IPCC default factors for coal, heavy fuel oil and diesel , [34] IPCC default factors for coal, heavy fuel oil and diesel , [34]																																											
Value applied	For coal, fuel oil and diesel as per plant in Table 99 <table> <tr> <th>Plant Name</th><th>Net calorific value coal (GJ/t)</th><th>Net calorific value fuel oil (GJ/t)</th></tr> <tr><td>Arnot</td><td>19.29</td><td>40.4</td></tr> <tr><td>Duvha</td><td>19.29</td><td>40.4</td></tr> <tr><td>Hendrina</td><td>19.29</td><td>40.4</td></tr> <tr><td>Kendal</td><td>19.29</td><td>40.4</td></tr> <tr><td>Kriel</td><td>19.29</td><td>40.4</td></tr> <tr><td>Lethabo</td><td>19.29</td><td>40.4</td></tr> <tr><td>Matimba</td><td>19.29</td><td>40.4</td></tr> <tr><td>Majuba</td><td>19.29</td><td>40.4</td></tr> <tr><td>Matla</td><td>19.29</td><td>40.4</td></tr> <tr><td>Tutuka</td><td>19.29</td><td>40.4</td></tr> <tr><td>Camden</td><td>19.29</td><td>40.4</td></tr> <tr><td>Grootvlei</td><td>19.29</td><td>40.4</td></tr> <tr><td>Komati</td><td>19.29</td><td>40.4</td></tr> </table>		Plant Name	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)	Arnot	19.29	40.4	Duvha	19.29	40.4	Hendrina	19.29	40.4	Kendal	19.29	40.4	Kriel	19.29	40.4	Lethabo	19.29	40.4	Matimba	19.29	40.4	Majuba	19.29	40.4	Matla	19.29	40.4	Tutuka	19.29	40.4	Camden	19.29	40.4	Grootvlei	19.29	40.4	Komati	19.29	40.4
Plant Name	Net calorific value coal (GJ/t)	Net calorific value fuel oil (GJ/t)																																										
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Camden	19.29	40.4																																										
Grootvlei	19.29	40.4																																										
Komati	19.29	40.4																																										

	Kusile	19.29	40.4
	Medupi	19.29	40.4
	Plant Name	Net calorific value Diesel (MJ/l)	
	Acacia	43	
	Port Rex	43	
	Ankerlig	43	
	Gourikwa	43	
Justification of choice of data or description of measurement methods and procedures applied	Tier 2 NCV (for South Africa) only available for coal. Tier 1 used for other NCVs		
Purpose of data	To calculate baseline emissions, using Equation 5.		
Comments	The data on fuel consumption and electricity generation is available for the selected m power plants, so Equation 4 under Option A1 from paragraph 49(a) in TOOL 07 is applied.		

Data / Parameter	AEG_{total}
Data unit	GWh
Description	Total electricity generation of all power plants directly supplying the system in year y
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website [14] for each power station.
Value applied	210 648
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above.
Purpose of data	To calculate baseline emissions
Comments	No additional comments

Data / Parameter	$SET_5 \text{ units}$												
Data unit	Dimensionless												
Description	In line with VT0011, the set of five power units, including power units registered as VCS or other GHG programs, that started to supply electricity to the grid most recently was identified												
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website [14] for each power station.												
Value applied	<table border="1"> <thead> <tr> <th>Plant name</th><th>Commissioning Date</th></tr> </thead> <tbody> <tr> <td>TWE Golden Valley Wind Power Project</td><td>2021/05/01</td></tr> <tr> <td>GS11949 Small-scale solar electrical programme, South Africa</td><td>2020/12/16</td></tr> <tr> <td>Solar Energy Programme for South Africa</td><td>2019/02/01</td></tr> <tr> <td>Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility</td><td>2017/10/31</td></tr> <tr> <td>Longyuan Mulilo De Aar 2 North Wind Energy Facility</td><td>2017/10/31</td></tr> </tbody> </table>	Plant name	Commissioning Date	TWE Golden Valley Wind Power Project	2021/05/01	GS11949 Small-scale solar electrical programme, South Africa	2020/12/16	Solar Energy Programme for South Africa	2019/02/01	Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31	Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31
Plant name	Commissioning Date												
TWE Golden Valley Wind Power Project	2021/05/01												
GS11949 Small-scale solar electrical programme, South Africa	2020/12/16												
Solar Energy Programme for South Africa	2019/02/01												
Longyuan Mulilo De Aar Maanhaarberg Wind Energy Facility	2017/10/31												
Longyuan Mulilo De Aar 2 North Wind Energy Facility	2017/10/31												
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table 10												
Purpose of data	To calculate baseline emissions												
Comments	This set of power units generate 0.78 % of AEG_{total} in 2024.												

Data / Parameter	$AEG_{set-5-units}$
Data unit	GWh
Description	Annual electricity generation of the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently, and that comprise 20% of AEG_{total}

Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and records published on Eskom's website [14] for each power station.
Value applied	1 643
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table above
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 0.78% of AEG_{total} in 2024.

Data / Parameter	$SET_{\geq 20\%}$
Data unit	Dimensionless
Description	The set of power units, including power units registered as CDM/VCS project activities, and that comprise 20% of AEG_{total}
Source of data	Determined from IPP database [35].

Value applied	No.	Plant name	Commissioning Date	Power Produced in Year: 2024 (MWh)
	Eskom Fleet Plants			
	1	Gourikwa	2007/07/01	2 038 316
	2	Medupi	2015/08/22	22 626 444
	3	Kusile	2017/08/30	10 262 515
	IPP Power Plants			
	4	TWE Golden Valley	2021/05/01	509 303
	5	GS11949 small scale solar electrical programme, South Africa	2020/12/16	22 500
	6	Solar energy programme for south africa	2019/02/01	381 930
	7	Longyuan Mulilo de Aar Maanhaarberg Wind Energy Facility	2017/10/31	290 500
	8	Longyuan Mulilo de Aar 2 North Wind Energy Facility	2017/10/31	439 600
	9	Green Power for South Africa	20/09/2011	1220
	10	Grid Connected Wind Power Plant in Witberg	2012/02/01	507
	11	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	375
	12	South Africa Wind Energy	01/01/2016	95
	13	West Coast 1 Wind Farm	01/06/2015	293
	14	Renewable Energy Carbon Program for Africa	01/07/2013	140
	15	Grid Connected Photovoltaic Renewable electricity generating facilities POA	19/07/2013	21
	16	South African Large scale Grid connected Solar Park Programme	05/04/2012	66
	17	Cookhouse wind farm in South Africa	01/12/2011	1057
	18	Red Cap Kouga Wind Farm	2011/11/02	290
	19	Karoo Renewable Energy Facility (Nobelsfontein Wind)	2012/12/01	878
	29	Karoo Renewable Energy Facility (Nobelsfontein Solar PV)	2014/04/01	95
	21	Bokpoort CSP (Concentrating Solar Power) Project, South Africa	2013/01/10	226
	22	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	130
	23	Electricawinds 30 MW Wind Project at Riverbank	2013/12/01	80
	24	Caledon Wind Farm in South Africa	2012/06/01	406
	25	Langa Energy Photovoltaic Solar Energy Facility	2012/03/05	140
	26	Indwe Wind Project	2012/06/01	148
	TOTAL			42 739 622
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table			
Purpose of data	To calculate baseline emissions			
Comments	This set of power units generate 20.29% of AEG _{total} in 2024.			

Data / Parameter	$AEG_{set \geq 20\%}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, including power units registered as CDM project activities, and excluding

	those commissioned more than 10 years ago, and that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and IPP database [35]
Value applied	42 739
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated 20.29% of AEG_{total} in 2024.

Data / Parameter	SET_{sample}
Data unit	Dimensionless
Description	The set of power units, (between $SET_{5\text{ units}}$ and $SET_{\geq 20\%}$) that comprises the larger annual electricity generation.
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 and records published on Eskom's website [14] for each power station to fulfil the Atmospheric emission license (EAL) requirements.

Value applied	No.	Plant name	Commissioning Date	Power Produced in Year: 2024 (MWh)
Justification of choice of data or description of measurement methods and procedures applied Purpose of data Comments	Eskom Fleet Plants			
	1	Gourikwa	2007/07/01	2 038 316
	2	Medupi	2015/08/22	22 626 444
	3	Kusile	2017/08/30	10 262 515
	IPP Power Plants			
	4	TWE Golden Valley	2021/05/01	509 303
	5	GS11949 small scale solar electrical programme, South Africa	2020/12/16	22 500
	6	Solar energy programme for south africa	2019/02/01	381 930
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	9	Green Power for South Africa	20/09/2011	1220
	10	Grid Connected Wind Power Plant in Witberg	2012/02/01	507
	11	Amakhala Emoyeni Grid Connected 138.6 MW Wind Farm, Phase 1	2013/03/01	375
	12	South Africa Wind Energy	01/01/2016	95
	13	West Coast 1 Wind Farm	01/06/2015	293
	14	Renewable Energy Carbon Program for Africa	01/07/2013	140
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	17	Cookhouse wind farm in South Africa	01/12/2011	1057
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	22	Springbok Grid Connected 55.5 MW Wind Farm (Longyuan Mulilo)	2013/06/01	130
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	24	Caledon Wind Farm in South Africa	2012/06/01	406
	25	Langa Energy Photovoltaic Solar Energy Facility	2012/03/05	140
	26	Indwe Wind Project	2012/06/01	148
		TOTAL		
	Eskom is South Africa’s national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in Table .			
	Table			
	To calculate baseline emissions			
	This set of power units generated 42 739 GWh in 2024, comprising 20.29% of AEG _{total} in 2024.			

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Data / Parameter	$AEG_{set\ sample}$
Data unit	GWh
Description	Annual electricity generation of the set of power units, including power units registered as CDM project activities that comprise 20% of AEG_{total}
Source of data	Determined from electricity generation data reported in Eskom's annual Integrated Report for 2024 [32] and IPP database [35]
Value applied	42 739
Justification of choice of data or description of measurement methods and procedures applied	Eskom is South Africa's national electricity utility, responsible for generating electricity, procuring renewable electricity from Independent Power Producers (IPPs) and for selling electricity to users. The data used is shown in
Purpose of data	To calculate baseline emissions
Comments	This set of power units generated >20% of AEG_{total} in 2024.

Data / Parameter	w_{BM}
Data unit	Percentage (%)
Description	Weighting of build margin emissions factor
Source of data	VT0011 Paragraph 86
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in VT0011
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the build margin emission factor is set at 50% for the first crediting period.

Data / Parameter	w_{OM}
Data unit	Percentage (%)
Description	Weighting of operating margin emissions factor
Source of data	VT0011 Paragraph 86
Value applied	50%
Justification of choice of data or description of measurement methods and procedures applied	Default value provided in VT0011
Purpose of data	To calculate baseline emissions, using Equation 7
Comments	Owing to the intermittent and non-dispatchable nature of solar power generation, the weighting of the operating margin emission factor is set at 50% for the first crediting period.

Data / Parameter	BE_y
Data unit	tCO ₂ e/yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 1

Value applied	Vintage period	Estimated baseline emissions (tCO ₂ e)
	19-NOV-2023 to 31-Dec-2023	76 226
	01-Jan-2024 to 31-Dec-2024	794 745
	01-Jan-2025 to 30-June-2025	395 778
	Total	1 266 749
Justification of choice of data or description of measurement methods and procedures applied	Equation provided in ACM0002.	
Purpose of data	To calculate baseline emissions, using Equation 1	
Comments	No additional comments	

6.2 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Source of data	Electricity reading meters
Description of measurement methods and procedures to be applied	Measurement data is managed by the project's SCADA system. The measuring and monitoring equipment is supplied by a combination of OEM suppliers, including Sneider Electric, Schweitzer Engineering Laboratories (SEL), Landis+Gyr and ABB,

	and was installed by competent instrumentation Technicians and OEMs.
Frequency of monitoring/recording	Continuous monitoring and monthly reporting
Value applied	Ex-ante values for the first crediting period: 870, 000 MWh /yr
Monitoring equipment	The measuring and monitoring equipment is supplied by a combination of OEM suppliers, including Sneider Electric, Schweitzer Engineering Laboratories (SEL), Landis+Gyr and ABB, and was installed by competent instrumentation Technicians and OEMs
QA/QC procedures to be applied	Metered electricity production will be recorded by the plant on the DCS system. This will be checked against the Eskom meter at the independent switching station. Any discrepancies will immediately trigger fault finding and corrective action procedures. There is also a backup meter both on site and at the switching station which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both Project personnel and Eskom personnel prior to billing and payment.
Purpose of data	Calculation of project emissions
Calculation method	Net electricity supplied to the grid by the project plant is directly measured and reported, not calculated.
Comments	No additional comments

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Project emissions in year y
Source of data	Quantified using Equation 8
Description of measurement methods	Quantified, not measured

and procedures to be applied	
Frequency of monitoring/recording	Monthly
Value applied	0
Monitoring equipment	Quantified, not monitored
QA/QC procedures to be applied	No additional QA/QC would be applied to this equation
Purpose of data	Calculation of project emissions
Calculation method	Equation 8
Comments	For all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected. This project does not involve geothermal power plants or water reservoirs of hydro power plants; therefore, project emissions for this renewable energy power generation will be zero.

Data / Parameter	ER_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Source of data	Quantified using Equation 9
Description of measurement methods and procedures to be applied	Quantified, not measured
Frequency of monitoring/recording	Monthly
Value applied	Ex-ante value is 810 840 per annum
Monitoring equipment	Quantified, not monitored

QA/QC procedures to be applied	No additional QA/QC would be applied to this equation
Purpose of data	Calculation of project emissions
Calculation method	Equation 9
Comments	The emission reductions determined at validation apply estimates of likely emissions reductions from the proposed project activity during the crediting period, using available ex-ante values. These will be revised during each verification using the actual data recorded (where applicable).

6.3 Monitoring Plan

Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters.

The facility has a TGS Prediktor Supervisory Control and Data Acquisition (SCADA) system that ensures plant-wide integration of devices and allows for effective plant operations and monitoring and reporting. Apart from reporting for this project, the data will also be used for reconciliation and checking the billing from the national utility for electricity supplied to the grid. All data and reports will be stored and backed up in accordance with the SCADA system design, this will be done off site and is linked to a Global Operations Centre (GOC) hub remotely. The GOC monitors numerous projects that the PP operates globally. The location of the meters on site is shown below. Figure 11 shows the location of the meters within the greater site layout, Figure 12 is a single line diagram (SLD) for SK1 for the metering configuration, Figure 13 is a photo of the SK2 meters within the Kenzeer 2 substation metering panel and Figure 14 is a line diagram of all 3 clusters (SK1, SK2 & Sk3) indicating the meter locations and within the Kenzeer substation complex and the feed to the Eskom Rooidam substation with the check meters location. The meters are connected to the SCADA system.

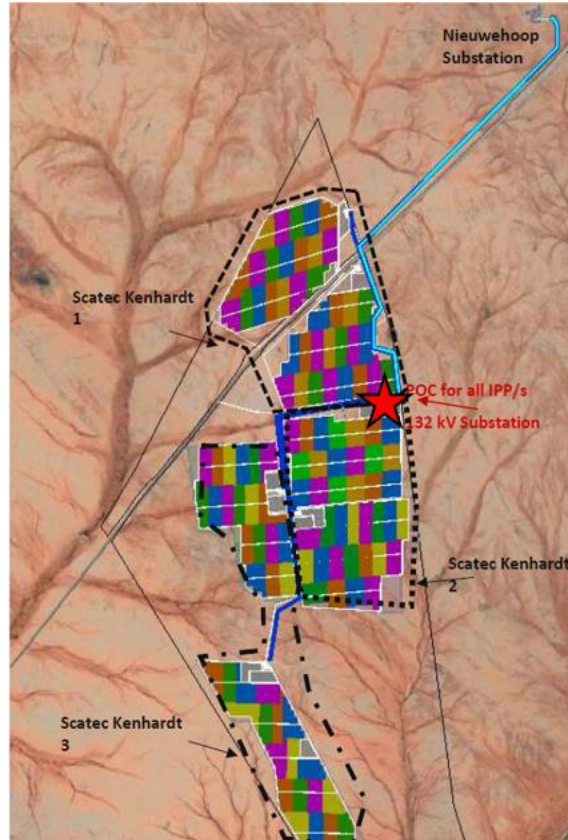


Figure 11: Meter locations on site (indicated by star shape)



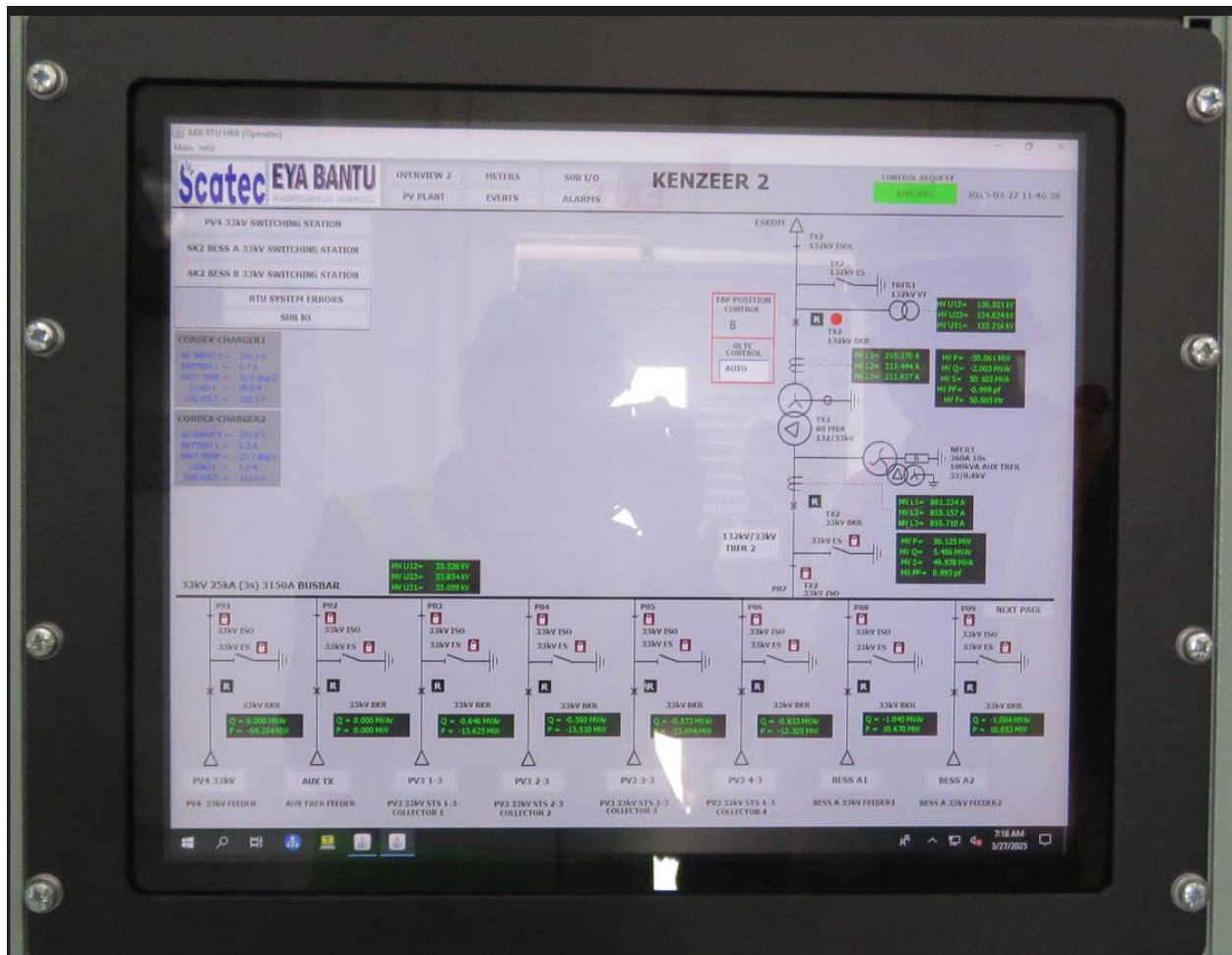


Figure 13: Indicative instrumentation SLD of metering configuration to 132kV

Scatec RMI Hybrid Metering Diagram

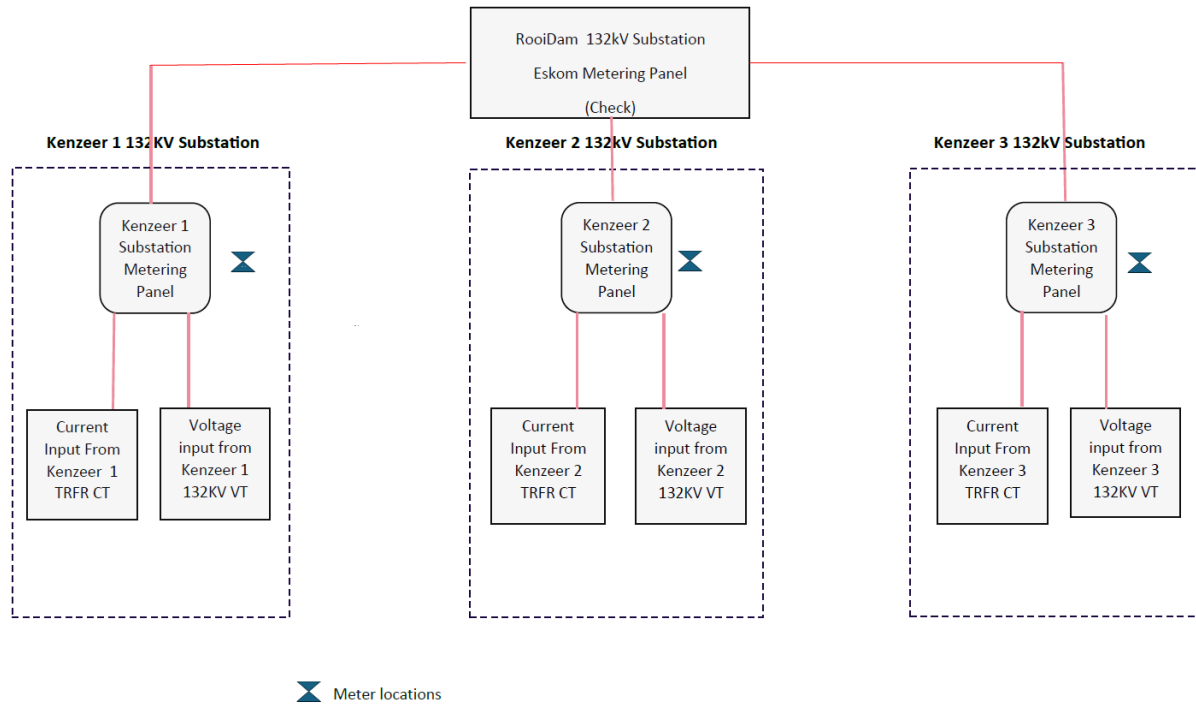


Figure 14: Line diagram of metering location and instrument panels feeding Rooidam

Measurement data is managed by the project's SCADA system. The measuring and monitoring equipment is supplied by a combination of OEM suppliers, including Sneider Electric, Schweitzer Engineering Laboratories (SEL), Landis+Gyr and ABB, and was installed by competent instrumentation Technicians and OEMs:

- Calibration under standard temperature and pressure conditions by OEM
- Installation and electrical connection by a certified external contractor
- Commissioning and final test under project conditions by the internal instrumentation team

All data is stored on a server on-site and also loaded onto a firewall protected cloud storage system. All measuring equipment strictly complies with their specified maintenance plans and calibration certificates are kept on record.

As per the applied methodology, electricity generation will be measured onsite at the outgoing switching station. This will be monitored on a continuous basis, with a monthly report being aggregated and reported upon. All monitoring and reporting will be done at the GOC unit in the Western Cape. The check meters, owned by the national utility (Eskom), is located at the adjacent Eskom Rooidam 132 kV Switching Station within the perimeter fence which serves as the grid tie-in point. Meters at both locations have a back-up meter installed as a failsafe.

The electrical meters on site, calibrated in accordance with the manufacturer's specifications, will form part of the planned maintenance schedule for both checking and calibration. Instrument accuracy checks and verification will be conducted by Competent Instrumentation technicians and records kept on site. Calibrations will be conducted by competent technicians qualified under the national SANAS accreditation. The applicable national standard to be applied is SANS 474 [36]. This defines the required frequency of calibration in §4.7.4 for meters greater than 10MVA as being every 5 years. In addition to the SANS 474 national standard, the Power Purchase Agreement (PPA) regulates the calibration requirement under Section 12 – Metering. These requirements have been incorporated into the planned operations and maintenance schedules of the facility. While the PPA does not stipulate a calibration frequency, and places no obligation on the national utility, the project has opted to provide a calibration frequency of a maximum of every 5 years as per the SANS474 national standard. The calibrations are carried out by competent and SANAS accredited third party instrumentation companies and calibration certificates are issued and kept on record.

Organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.

Data collection, consolidation and results analysis will be undertaken by a team who are adequately trained and competent. They will be responsible for the maintenance (planned and unplanned), checking and arranging calibration for plant equipment. They will also be responsible for maintaining records of calibrations and procurement of any instrumentation equipment that may be required for the operation of the plant. All instrument technicians will be suitably competent and records of their training and qualifications will be maintained by the Human Resources department. Only SANAS qualified organizations will conduct calibrations. These organizations will be part of an approved suppliers list and are also integrated into the certified ISO9001, ISO14001 and ISO45001 management systems.

Procedures for internal auditing and QA/QC

Metered electricity production will be recorded by the plant on the SCADA system. This will be checked against the Eskom meter at the independent Rooidam switching station. There is also a backup meter both on site and at the Rooidam switching station which are used as an additional failsafe for QA/QC purposes. All reported electricity generation values will be checked by both Scatec personnel and Eskom personnel prior to billing and payment. The final reporting checks are conducted off site at the Scatec GOC based in Cape Town.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	$EG_{facility,y}$
Data unit	MWh/Year
Description	Quantity of net electricity generation supplied by the project to the grid in year y
Value applied:	1 359 173
Comments	From Plant reporting

Data / Parameter	ER_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Value applied:	1 266 749
Comments	Calculated from Equation 1

Data / Parameter	PE_y
Data unit	tCO ₂ /yr
Description	Emission reductions in year y
Value applied:	0

Comments	As per methodology
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Data / Parameter	$EF_{grid,CM,y}$
Data unit	tCO ₂ /MWh
Description	Combine margin CO ₂ emission factor in year y
Value applied:	0.932
Comments:	Calculated above

7.2 Baseline Emissions

The baseline emissions were quantified in accordance with the methodology ACM0002 version 22.0. For this project, it includes electricity delivered to the South African National grid. This was calculated as per the calculation steps described below.

Baseline Emission Calculation

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Equation 8 (ACM0002 Equation 11)

Where:

BE_y = Baseline emissions in year y (tCO₂/yr)

$EG_{PJ,y}$ = Net electricity fed into the grid supplied by the project activity in year y
(MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in

year y calculated using TOOL07 (t CO₂/MWh)

The results obtained are: 1 266 740 t CO₂/ annum, as shown below:

Parameter name	Value	Unit
BE_y	1 266 740	t CO ₂ / yr
$EG_{PJ,y}$	1 359 173	MWh/yr
$EF_{grid,CM,y}$	0.932	t CO ₂ /MWh

7.3 Project Emissions

As per the approved consolidated Methodology ACM0002, “Grid-connected electricity generation from renewable sources” (Version 22.0) para 40: “For most renewable energy power generation project activities, PE_y = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y} + PE_{PSP,y}$$

Where,

PE_y = Project emissions in year y (t CO₂e/yr)

PE_{FF,y} = Project emissions from fossil fuel consumption in year y (tCO₂/yr)

PE_{GP,y} = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (tCO₂e/yr)

PE_{HP,y} = Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)

PE_{BESS,y} = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (tCO₂e/yr)

PE_{PSP,y} = Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t CO₂e/yr)

As the project activity is the installation of a new grid-connected solar Power plant and BESS technology with a hybrid power controller and does not involve any project emissions from fossil fuel, operation of dry, flash steam or binary geothermal power plants, and from water reservoirs of hydro power plants. Therefore PE_{FF,y}, PE_{GP,y}, PE_{HP,y}, PE_{BESS,y}, PE_{PSP,y} are equal to zero and thus, PE_y = 0.

7.4 Leakage Emissions

As per the applied methodology, no other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g.: extraction, processing, transport etc.) are neglected.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
19-NOV-2023 to 31-DEC-2023	76 226	0	0	76 226	0	76 226
01-JAN-2024 to 31-DEC-2024	794 745	0	0	794 745	0	794 745
01-JAN-2025 to 30 JUN-2025	395 778	0	0	395 778	0	395 778
Total	1 266 749	0	0	1 266 749	0	1 266 749

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
19-Nov-2023 to 31-Dec-2025	87 841	76 226	13%	Phased approach during plant start-up and initial troubleshooting
01-Jan-2024 to 31-Dec-2024	810 840	794 745	2%	Planned and unplanned Maintenance
01-Jan-2025 to 30-Jun-2025	405 420	395 778	1%	Planned and unplanned Maintenance

Total	1 304 101	1 266 749	3%	Startup troubleshooting and Planned and unplanned maintenance
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APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification
	No Commercially sensitive information was included in this version of the PD	

APPENDIX 2: BIBLIOGRAPHY

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